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Computation offloading in the edge-to-cloud compute continuum: a survey of federated architectural solutions

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Abstract
Computation offloading involves transferring computation to resourceful nodes to overcome resource limitations, particularly within the emerging federated edge-to-cloud computing infrastructures. The growth of IoT applications has hugely increased the need for effective and decentralised offloading strategies considering the highly dynamic and dispersed edge-to-cloud compute continuum. Recently, researchers have used various implementation techniques—ranging from rule-based systems, heuristic, and machine learning-based intelligence—to investigate issues related to offloading, including edge computing, fog computing, and edge-to-cloud compute continuum. Although several review papers are available that provide a comprehensive analysis of existing research works on computation offloading, most surveys have largely overlooked the specific challenges and characteristics of federated and distributed edge-to-cloud execution models. In contrast, this paper aims to compile and synthesise computation offloading research with a specific focus on the federated edge-to-cloud ecosystem and distributed execution solutions. We first propose a detailed taxonomy focused on the decentralisation aspect and federated coordination. This taxonomy is then used as a unified framework to critically review the existing research landscape. Finally, we identify and discuss the key challenges that require further attention, providing insights to guide future developments in distributed, federated offloading systems.

Keywords Edge computing · Cloud computing · Federation · Distributed offloading · Optimization · Machine learning

1 Introduction

In today's interconnected world, the growth of the Internet of Things (IoT) has given rise to a new breed of application that, in addition to using cloud resources, also depends on IoT devices to collect and perhaps analyse data from the local environment. Smart manufacturing, smart cities, augmented reality, and self-driving vehicles are just a few of the many fields in which the use of such systems has been expanding quickly and is now becoming a necessary component of our everyday life. To achieve low-latency analytics, data security & privacy in a distributed environment, context awareness, and concurrent access to geographically distributed heterogeneous computational resources, a traditional cloud-only computing architecture is often impractical if not inadequate spullah2021micado. As a result, new computational paradigms like edge computing, fog computing and edge-to-cloud compute continuum have emerged to overcome the aforementioned challenges.

The terms edge and fog computing are often used interchangeably to loosely refer to moving computation away from the central cloud to nodes closer to endpoints at the network edge. They both aim to reduce the amount of data sent to the cloud and achieve lower latency, however, there are subtle differences between the two. For example, with reference to Fig. 1, the nodes between the cloud and the IoT sensors and actuators are represented by the fog computing layer, which sits between the cloud and the edge and may span several layers of the network topology [1]. In contrast, edge computing uses nodes that are either integrated within

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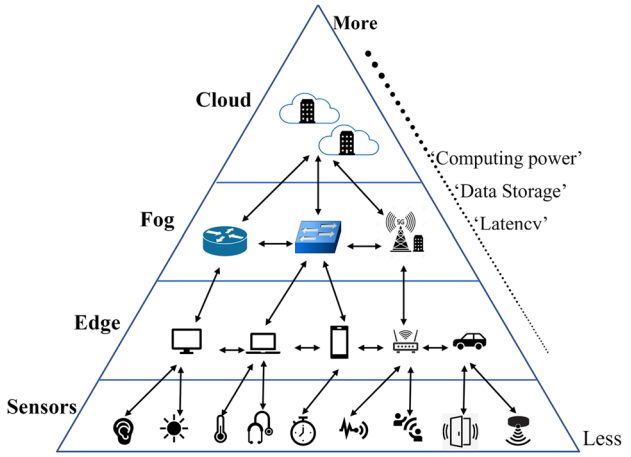


Fig. 1 A generic architecture for edge-to-cloud environments adopted from [8]

the connected device or very close to the IoT devices in terms of network proximity, sometimes just one or a few hops away [2].

The edge-to-cloud compute continuum—also known by other names such as cloud continuum, cloud-edge continuum, or cloud-to-things continuum—refers to the extension of cloud with energy-efficient and low-latency devices closer to the data sources located at the network edge [3, 4]. More specifically, it extends the traditional Cloud towards multiple entities such as Fog, Edge, and IoT to provide different capabilities including analysis, processing, storage, and data generation [5]. Thus, it achieves better performance overall in regard to computation and transmission in comparison with the cloud-only or edge computing model [6].

The federated edge-to-cloud continuum architecture is conceptualised as a multi-layered system with three main layers. The first layer is the edge layer, which consists of IoT/edge devices, generating tasks and conducting initial processing with limited capacity and resources. The second layer is a fog layer, in which computational nodes located closer to data sources perform low-latency processing, aggregation, and context-aware services. The third layer is a cloud layer providing large-scale storage, advanced analytics, and global orchestration for high-resource-demand tasks. In this federated model, resources at these three layers are dynamically coordinated to compute local real-time tasks with the edge devices, aggregate at fog nodes, and selectively transfer to the cloud layer for further processing. This hierarchical structure optimises resource utilisation, enhances privacy, reduces latency, and scalability to support developing applications across smart cities, healthcare, industry 4.0, and beyond [7].

In the edge-to-cloud compute continuum, a computational task can be scheduled to execute either on a user device, an edge or a cloud. However, at times, due to resource limitations (e.g. insufficient capacity, low battery power) and/or various associated optimisation objectives (e.g. to achieve low latency), the computational task is required to be moved to a different resource. This process of migrating computational tasks is generally referred to as computation offloading. Over the past several years, the growth of IoT applications has hugely increased the need for effective computation offloading strategies to address the inherent complexity of the highly dynamic and dispersed edge-to-cloud compute continuum.

Computation offloading plays a crucial role in minimising latency and ensuring QoS objectives. For example, tasks may be offloaded from cloud to edge devices in order to balance system overheads, energy consumption, and system performance, alternatively, when edge devices do not have the processing capability, the computation can be migrated to a nearby edge device or to the cloud to avoid impact on overall system performance [9]. Computation offloading is a complex task. There are many different aspects that are to be considered during an offloading process. For example, the overall key purpose such as performance improvement and/or energy reduction, the tradeoff between computation vs. communication, and the decision of when, where and how to offload.

To deal with the complexity and challenges of computation offloading, researchers have used a variety of implementation techniques—ranging from rule-based systems, heuristic, optimise and machine learning-based intelligent approaches—to investigate issues related to offloading in a variety of different paradigms, including edge computing, fog computing, mobile edge computing, and edge-to-cloud [10–13]. Additionally, there are also several review papers available that provide a comprehensive analysis of computation offloading literature from different viewpoints. For example, the authors in [14–17] delved into detailed implementation techniques, papers [18, 19] focused on the underlying objective or decision-making criteria-based, and the authors in [20, 21] considered a paradigm wide scope. More details of these review papers are discussed in Sect. 2.

In contrast to existing review papers, we are interested in reviewing the existing offloading solutions based on their control topology—dissemination of decision-making nodes that control the computing infrastructure and offloading decisions—and their overall support for the essential characteristics required for the development of an offloading mechanism. Based on control topology (or execution model), the overall landscape of offloading solutions can be classified mainly into two categories, i.e., centralised or decentralised.

In a centralised approach, a central entity collects data from the entire compute continuum and make system

decision. Overall, it offers the benefits of consistent decision-making, thanks to the fact that all relevant information is located in a single place, and ease of implementation. However, such approaches are also criticized for lack of scalability, single point of failure, and an easy central target for cyber-attacks [22]. Furthermore, the central decision-making system can easily become a bottleneck from an efficiency viewpoint, especially, considering the distributed nature of the dynamic compute continuum, a centralised model fosters serious concerns such as the continuous transfer of recurrent monitoring data to a central point from distributed resources, data privacy and security, unnecessary communication overhead, and unreliable connectivity.

A decentralised approach on the other hand consists of multiple interconnected entities to control instead of a single entity as in the case of centralised systems [23]. A decentralised execution model aims to address the issues highlighted in the case of centralised approaches. In reference to this aspect, the scope of our review paper is to review offloading decisions that follow a decentralised execution approach with a coverage centred around the federated cloud-to-edge environment, distinguishing it from earlier reviews that focused on specific environments like MEC [21], MCC [20], etc. More specifically, the key contributions of this review paper are as follows:

- We performed a comprehensive analysis and identified a wide range of fundamental attributes associated with offloading from a decentralised viewpoint in the edge-to-cloud compute continuum. These identified characteristics serve as integral components and hold significant importance in the evaluation of computation offloading solutions. Using these attributes, we put forth an innovative taxonomy, which serves as a unified framework and plays a crucial role in comprehending and analysing the current landscape of decentralised offloading solutions.
- A comprehensive examination in light of the proposed taxonomy was conducted, encompassing a diverse array of computation offloading solutions aimed at the edge-to-cloud compute continuum.
- We present a summary of the available experimental environments and benchmarks that are/can be used in the research of decentralised offloading solutions.
- Lastly, we list and discuss the open challenges associated with decentralised offloading solutions that require future attention.

The remainder of this paper is structured as follows: Sect. 2 reviews existing relevant survey papers, highlighting their key strengths and weaknesses, and justifying the need for yet another review on computation offloading. Section 3 presents the research methodology that is adopted to carry out this review work. Section 4 discussed the proposed taxonomy in detail, whereas Sect. 5 discusses the existing offloading solutions in light of the proposed taxonomy and presents the results. Section 6 outlines open issues and highlights future research directions. Lastly, Sect. 7 concludes the paper.

2 Related work

This section discusses recent studies that survey computation offloading. The objective is to thoroughly analyse their merits and weaknesses while emphasising the distinctions they exhibit when compared to the review conducted in this study. In Table 1, all related surveys on computation offloading summarised.

Several review papers specifically focused on the underlying implementation techniques. For example, Lin et al. [15] and Sadatdiynov et al. [14] reviewed the existing related literature by classifying them into different modelling approaches such as (non-) convex optimization, Markov Decision Process (MDP), Lyapunov optimization, game theory, and machine learning. Similarly, Shakarami et al. [24] focused on the stochastic-based offloading approaches, where they classified the existing literature into different modelling categories such as Markov chain, hidden Markov, semi-Markov-based models, etc. In addition to covering state-of-the-art developments, Peng et al. [25] explored the core components of Deep Reinforcement Learning (DRL) algorithm design with an emphasis on the target environmental factors, Markov Decision Process (MDP) model construction, and improved learning strategies. Kar et al. [26] and Saeik et al. [11], both have focused on optimisation and machine learning techniques. Saeik et al. [11] also cover solutions that are based on control theoretical-based approaches.

Shakarami et al. [27] also reviewed machine learning-based offloading mechanisms, where they broadly categorized the reviewed solutions within three categories, including supervised, unsupervised, and reinforcement learning. Le et al. [28] reviewed the existing related literature by classifying them into two categories, such as distributed learning for IoT applications and IoT services, each of them has four subcategories, including unsupervised learning, supervised, semi-supervised, and reinforcement learning. Feng et al. [16] covered a large range of implementation aspects, including heuristic algorithms, Lyapunov optimization, game theory, Markov Decision Process (MDP), and Reinforcement Learning (RL). Zaman et al. [29] focused on the review of mobility-aware computational offloading approaches, where they classified the relevant solutions based on different kinds of models including

Table 1 Comparison table of existing surveys on computation offloading

Reference	Coverage		Years (up-to)	Type		Article	Contribution
	Single layer	Continuum		Selection	Classified		
[15]	✓		2019	✓		243	Lyapunov optimization, game theory, machine learning
[14]	✓		2022		✓	98	Lyapunov, heuristic, game theory, ML
[24]		✓	2020		✓	150	Stochastic-based offloading mechanisms
[25]	✓		2023	✓		204	Deep Reinforcement Learning (DRL)
[11]		✓	2021		✓	249	Mathematical, artificial intelligence, control theory
[26]	✓		2022		✓	152	Machine learning
[27]	✓		2020		✓	123	Machine learning
[28]		✓	2023		✓	310	Distributed machine learning
[16]		✓	2021		✓	190	Heuristic, Lyapunov, GT, Markov Decision Process, RL
[29]	✓		2021		✓	120	Mobility-aware computation
[30]	✓		2017		✓	60	Mobile data offloading
[31]	✓		2019	✓		107	Gaming theory, heuristics based
[32]		✓	2022		✓	181	Meta-heuristic, machine learning, hybrid algorithms
[33]	✓		2022		✓	196	Lyapunov, ML, game theoretic, heuristic techniques
[34]	✓		2020		✓	244	Vehicular computing
[20]		✓	2015	✓		42	Smart mobile devices (SMDs), framework mechanisms
[18]		✓	2023		✓	215	Markov decision process, Q-learning, MCC, Cloudlet, MEC
[21]		✓	2022		✓	84	MCC, Cloudlet, MEC, 5G network, Machine learning
[35]		✓	2020		✓	158	Decentralised task offloading, Edge-cloud computing
[36]	✓		2019	✓		224	Heterogeneous computing, Fuzzy control, Genetic, Markov

software-defined networking (SDN) model, Stochastic models, Random walk model, User movement tracking (UMT) based model, Machine learning, etc. Lastly, the focus in [30] includes offloading models, such as the channel model, computation and communication model, and energy harvesting model.

In contrast to the above-mentioned implementation-specific review papers, there are some review papers that follow an overall objective-specific approach. For example, Jiang et al. [31] reviewed the existing landscape from the overall objective of the underlying solution considering aspects like energy consumption minimization, QoS guarantee, and Quality of Experience enhancement. They also reviewed different resource scheduling approaches, as well as discussed, the key phenomenon of computation vs. communication considering the overhead of computation offloading decision making. Mahapatra et al. [32] also considered a single computing paradigm and examined task scheduling and/or load balancing algorithms to satisfy various QoS parameters, whereas this paper considers a wide range of computing paradigms, beginning with the Cloud computing layer at the top and ending with the IoT layer at the bottom. Similarly, Kumari et al. [33] focused on the underlying decision-making factors such as network architecture and optimization metrics.

There are also some domain-specific review papers. For example, the scope of De Souza et al. [34] only covers solutions from vehicular environments rather than the entire landscape of computation offloading. Akherfi et al. [20] focused on solutions proposed for computation offloading in Mobile Cloud Computing (MCC), specifically focusing on mobile computing and Cloud Computing (CC) approaches. They also present an overview of current offloading frameworks and computation offloading techniques, which conducted an analysis of these approaches, including an examination of their main critical issues. Dong et al. [18] concentrate on solutions proposed for task offloading in Mobile Edge Computing (MEC) environments, specifically focusing on delay and bandwidth in 5G technology. They also present an overview of current offloading aspects, including reducing delay, minimising energy consumption, balancing energy consumption and delay, enabling high-computing offloading, and addressing different application scenarios. Similarly, Mustafa et al. [21] targeted Mobile Cloud Computing (MCC) and Mobile Edge Computing (MEC) environments, where the review focused on aspects like offloading granularities, underlying implementation techniques, resources nature, optimisation scenarios, and offloading applications.

Lastly, there are several review papers that target the computation offloading landscape in the federated edge-to-cloud ecosystems. For example, the scope of Wang et al. [35] includes offloading approaches in the edge-to-cloud continuum. Their review is performed in light of taxonomy that is based on the overall high-level characteristic of offloading approaches. Analogously, Nguyen et al. [36] also focused on solutions proposed for computation offloading in heterogeneous edge-to-cloud computing, specifically focusing on smartphone applications based on three

models, called stochastic geometry, parallel computing, and queueing theory technique. Zheng et al. [13] presented a high-level overview of computation offloading, where they covered aspects like various offloading scenarios, factors influencing the offloading process, and diverse offloading strategies. The scope of this paper, however, is limited to an edge-only model.

In contrast to all the above-mentioned papers, our paper has several unique aspects. For example, unlike most previous review papers that delved into detailed implementation techniques such as [14–16, 29], our review concentrates on high-level characteristics of the offloading mechanism. Furthermore, unlike focusing on only the underlying objective or decision-making criteria-based review papers (e.g., [21, 31, 34]), our paper considers this as an essential characteristic of offloading solutions rather than a classification criterion. Lastly, and more specifically, the scope of this paper is to review offloading decisions that follow a decentralised execution approach with a coverage centred around the federated edge-to-cloud environment, distinguishing it from earlier reviews that focused on specific environments like MEC, MCC, etc. In terms of the overall approach, our review paper follows the model applied by Wang et al. [35], where the review is carried out using a detailed taxonomy. Our paper follows a similar model, where the proposed taxonomy in our paper complement and substantially extend Wang et al. taxonomy, however, the scope of our paper is to cover a more in-depth analysis of decentralized offloading approaches.

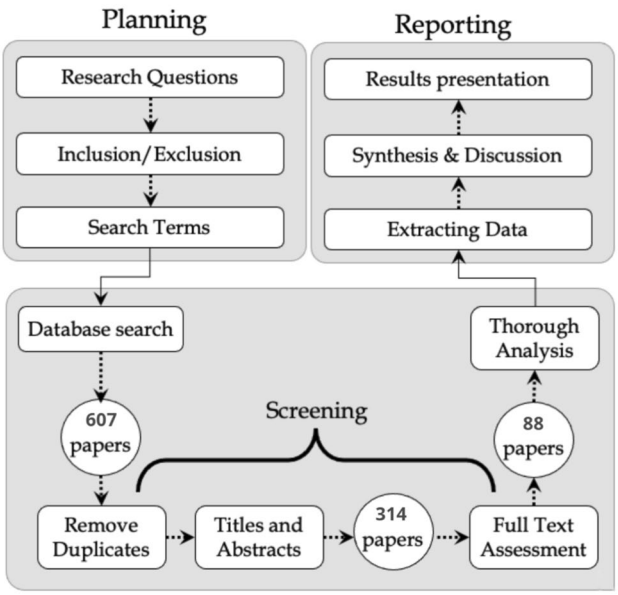


Fig. 2 The employed research methodology

3 Research methodology

This section discusses the overall research methodology employed to carry out the review. We adopted a systematic literature review approach that follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [37, 38]. PRISMA provides a structured framework for identifying, screening, and selecting relevant studies in a transparent and reproducible manner. In our review, all candidate solutions as well as the different aspects of decentralised offloading in the cloud-to-edge continuum can be thoroughly captured and examined. Overall, our methodology (see Fig. 2) mainly consists of the following three stages:

3.1 Planning the review

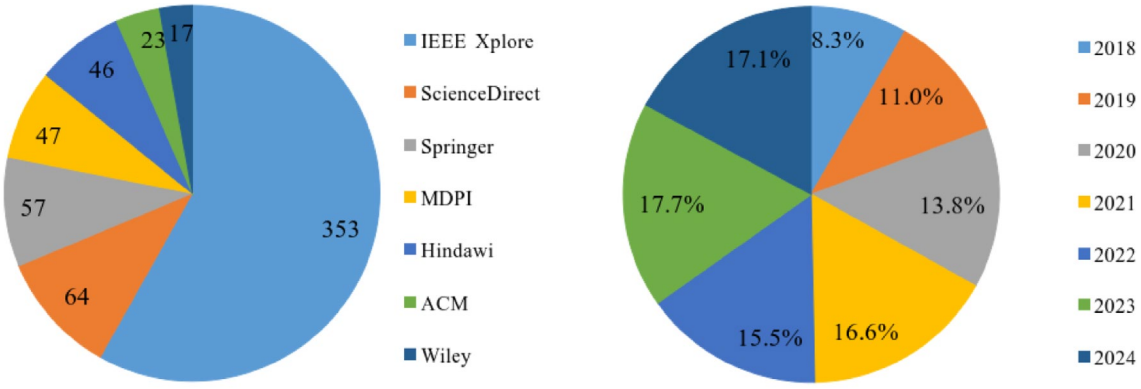
The planning stage consists of three important aspects including: (1) the definition of research questions (RQs) that set the foundation for the review, (2) setting up inclusion/exclusion criteria for selecting the related papers, and lastly, 2) the definition of search criteria considering the scope of the paper. The following RQs are formulated to guide this review:

1. What are the popular decentralised architectures, frameworks, and models of computation offloading, and how do they perform in terms of performance, scalability, and ease of deployment?
2. What are the key performance metrics used to evaluate and compare different computation offloading solutions?
3. What benchmarks, simulations, and tools currently exist to support the evaluation of the computational offloading mechanism?
4. What are the promising future research directions to address the open research challenges and issues related to decentralised architecture-based offloading solutions?

The above-mentioned RQs led us to the development of the proposed taxonomy that captures a large range of essential characteristics and features related to offloading in an edge-to-cloud continuum with a focus on decentralisation. This taxonomy is further used as a unified framework to critically review the landscape of existing research work.

Next, the inclusion/exclusion criteria considered the following four key aspects:

1. Relevance: We only considered those studies that explicitly proposed computation offloading solutions in the context of an edge-to-cloud ecosystem with a focus on decentralised control methodology.



(a) Database specific (b) Yearly

Fig. 3 Initial search results

2. Sources: We primarily considered peer-reviewed articles, book chapters, and conference/workshop proceedings.
3. Study design: Both empirical and theoretical studies, including case studies, simulation, and reviews that provide insights into the architectures, protocols, challenges and benefits related to computation offloading, are considered.
4. Date range: To encompass recent research works in the field, we included studies that were published in the last seven years, i.e. between 2018 and 2024.

Lastly, the definition of search criteria aims to identify the terms that are used to identify the relevant studies. Our search terms for this review are as follows:

Continuum

AND

(Computation OR Task OR Workload)

AND

(Transfer OR Migration OR Offloading)

The above-mentioned search criteria consist of three parts connected through the 'AND' primitive. The first part represents the target compute continuum domain. However, as the compute continuum is referred to with different names by different authors [5], we used all different variations to avoid missing relevant research work. These variations include terms such as Cloud-to-Edge, Edge-to-Cloud, Cloud-to-Things, Cloud-Fog-Edge, Edge-Fog-Cloud, and Cloud-Edge. The second part represents the different variations of the term 'computation', whereas the last part represents some of the variations for the term 'offloading'. These search terms aim to encompass a broad range of high-quality research articles, ensuring comprehensive coverage of the relevant subject matter.

3.2 Conducting the review

The review is carried out at this stage. It involves the execution of the following three steps including, performing the search, screening process to identify studies relevant to the scope of the paper, and performing a thorough analysis of the selected studies.

For the search, we used the search terms from Sect. 3.1 and looked for the relevant studies in the popular research databases including IEEE Xplore[1], Springer[2], ScienceDirect[3], ACM Digital Library[4], MDPI[5], Wiley[6], Hindawi[7]. Based on our search terms, we identified a total of 607 papers returned as a result of our search. Figure 3a presents the database-specific distribution of the obtained result, whereas Fig. 3b shows the yearly distribution of the obtained result.

On the one hand, as indicated in Fig. 3a, a substantial portion, approximately 353 papers, representing more than half a percent of the total relevant papers, has been published in IEEE Xplore. In the second tier, ScienceDirect has contributed around 64 papers, while Springer, positioned in the third tier, has published 57 papers. Furthermore, both

[1] https://ieeexplore.ieee.org/.
[2] https://link.springer.com/.
[3] https://www.sciencedirect.com/.
[4] https://dl.acm.org/.
[5] https://www.mdpi.com.
[6] http://onlinelibrary.wiley.com.
[7] https://hindawi.com.

MDPI and Hindawi have each contributed approximately 46 papers. Finally, ACM and Wiley, with 23 and 17 papers respectively, are among the publishers with fewer contributions in this context.

Figure 3b illustrates the entirety of relevant papers published between 2018 and the end of 2024. Notably, 34.8 percent of these relevant papers were published in the year 2023 and 2024, with roughly 17 percent in each year. Additionally, there is roughly a 16 percent overlap in papers published each year, 2021 and 2022. Furthermore, 24.8 percent of the entire set of papers were published in the two years, 2018 and 2019. Lastly, only 8.3 percent of all the relevant papers were published in the initial year, 2018.

The second step involves the screening of the studies found in the initial search. In the screening process, firstly, we removed any duplicates, followed by the screening of titles and abstracts to filter out studies that are not relevant to the scope of this paper. Furthermore, all review papers in the scope of offloading have been removed. Both these steps helped us in narrowing down the overall selection to 314 papers. These papers are the ones that warrant a full-text assessment for the comprehensive evaluation and to determine the relevance and value of the undertaken research. As a result of this step, our final set of studies reached 88 papers for thorough analysis. In this stage, the full text of the articles is thoroughly examined to ensure their quality and reliability. We adopted a holistic approach that focuses on the overall concept of the paper, encompassing broad characteristics of the offloading mechanism and evaluation within the context of decentralisation aspects, rather than delving into intricate implementation techniques.

3.3 Reporting the review

This final stage of the methodology involves extracting relevant data extraction, synthesising and reporting results. Both qualitative and quantitative approaches are adopted. From a qualitative viewpoint, the different solutions are analysed and discussed by clustering them into different related categories.

4 Taxonomy

Figure 4 presents our proposed taxonomy. The key purpose of this taxonomy is to have a unified framework that assists us in conducting a thorough review of the available computation offloading approaches. For this taxonomy, we have identified a diverse array of attributes associated with the offloading of IoT applications in the edge-to-cloud compute continuum based on an extensive analysis of the studies discussed in Sect. 2. These attributes are essential for the implementation of an offloading solution. These attributes are mainly distributed into the following four categories.

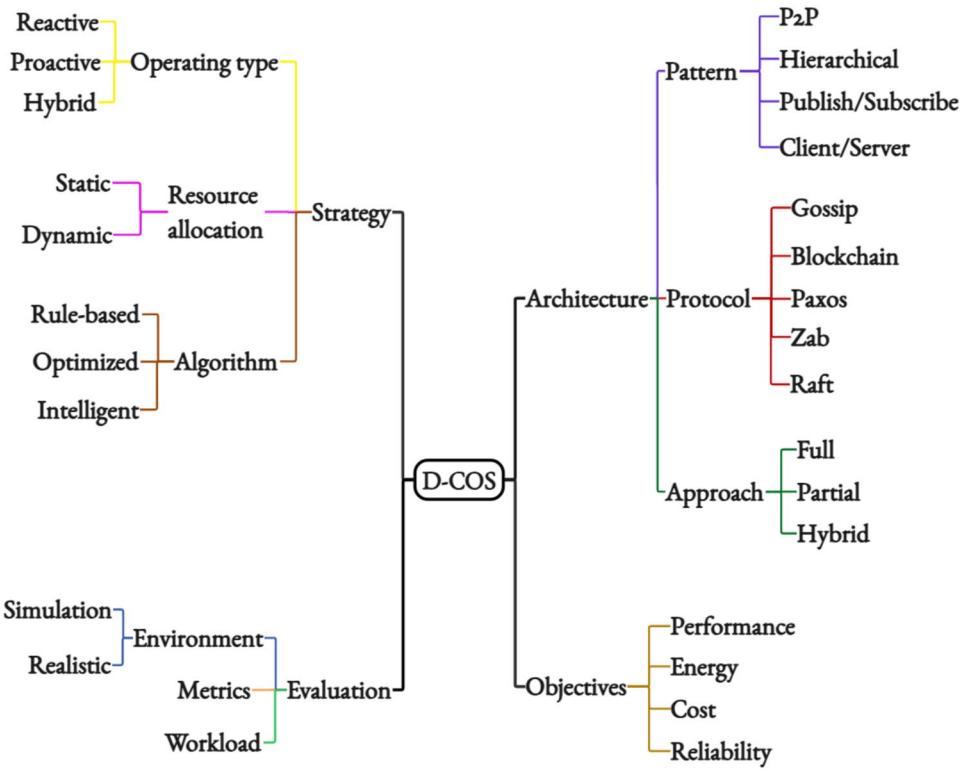


Fig. 4 Decentralised Computational offloading solutions (D-COS)

4.1 Architecture

The architecture of an offloading solution determines how the overall system operates to perform key functions, such as collecting information and decision-making. It plays a crucial role in terms of overall performance, scalability, resilience, and responsiveness. To delve deeper into the architectural details from a decentralise viewpoint, this category is further divided into the following key aspects:

- **Pattern** refers to the distributed pattern adopted for the implementation. Some such commonly adopted patterns include peer-to-peer (P2P), hierarchical, publish/subscribe, and Client/server. P2P is a decentralised networking model where each computing node acts as both a client and a server. It allows direct data sharing between nodes for file sharing, distributed computing, and real-time communication. Hierarchical patterns use a tree-like structure with a central node that controls the network and delegates tasks to other nodes. Publish/subscribe is a messaging pattern where publishers send messages to subscribers, which is commonly used for event notification, real-time communication, and streaming data. Additionally, in the client/server pattern, clients send requests to servers, and servers respond [39].
- **Protocol** captures the underlying distributed protocol used for the implementation of reviewed offloading solutions. Some of the commonly used protocols include Gossip, Paxos, Blockchain, Zab, and Raft. In gossip protocol, nodes periodically select a few peers to exchange information with, and the exchanged information is then further spread to other nodes by the receiving nodes. Paxos is a consensus algorithm designed to ensure agreement among a group of distributed nodes or processes in a computer system. Blockchain is a distributed ledger for secure and immutable transaction records, commonly applied in financial transactions and supply chain management. Raft and Zab are consensus protocols similar to Paxos, with Raft preferred for simpler implementations, while Zab is designed for asynchronous systems with intermittent communication in applications like distributed storage and real-time streaming [40].
- **Approach** determines the type of the applied computation offloading technique. This can be one of the following three types: full, partial, or hybrid. In the full approach, the offloaded computation task can be treated as an ”all or none” migration, which can be either executed locally or remotely [24]. The partial approach allows the division of a computation task, where the critical portion of the task is selected and offloaded to a remote location, while the remaining parts are executed locally [41]. The hybrid approach, on the other hand, is the one that support both full as well as partial offloading.

4.2 Strategy

This category focuses on the dynamics of the offloading mechanism itself. We recognize that any offloading mechanism is attributed to how it operates, the nature of resource allocation, and how it is implemented. Based on these aspects, we define strategy through the following three key attributes:

- **Operating type** represents the underlying nature of decision-making in a given offloading mechanism. This can be of the following three types: reactive, proactive, and hybrid. A reactive scheme is one where offloading takes place in response to the happening of some events, e.g. the sudden increases in network traffic or significant reductions in battery power of a device. A proactive strategy, on the other hand, anticipates future states and makes offloading decisions in advance [42, 43]. Lastly, the hybrid offloading model includes both reactive and proactive behavior [44].
- **Resource allocation** entails the allocation of resources like CPU for the offloaded tasks. Additionally, it involves the identification of nodes that help in the execution of local computing tasks. Generally, resource allocation in computation offloading can be mainly divided into two categories: static and dynamic. Static is referred to as the resource allocation strategy, where the offloading decisions are predetermined before task execution and remain fixed throughout the task execution life cycle. Dynamic resource allocation, on the other hand, involves real-time monitoring of wireless and edge-to-cloud computing resources, requiring adjustments in the execution strategy during the life cycle of tasks.
- **Algorithm** refers to the underlying computational approaches that are used for the implementation of an offloading mechanism. They encompass a range of computational methods, however, we generalise these mainly into the following three categories: (A) Rule-based offers application owners to provide predefined rules and logic to perform the offloading function; (B) Optimised includes all those approaches that aim to find optimal solutions amongst the many different possibilities; and (C) Intelligent includes the machine learning-based approaches, where the system learns patterns from available data and makes informed resource management decisions. Each of the above-mentioned approaches offers unique characteristics and can be suited for different scenarios. For example, the Rule-based systems are often fast and transparent, however, offer

very limited or no flexibility [45]. Optimised solutions, on the other hand, are praised for their production of optimal solutions, however, they are also criticised for being computationally expensive [45]. Lastly, intelligent approaches are adaptive and offer flexibility [46], however, its performance hugely depends on the availability and quality of training data.

4.3 Objectives

This attribute refers to the target objective of every offloading solution aims to optimise one or multiple specific objectives. We mainly categorize the possible objectives into the following four groups:

- **Performance** is the most commonly employed objective that deals with the improvement in the overall efficiency of the system in relation to various aspects such as service latency, network bandwidth, resource utilization, and/or workload distribution [40, 47].
- **Energy consumption** is another common objective target by the offloading solutions. This could include various energy-related aspects, e.g. enhancing battery longevity by mitigating energy consumption [48, 49]. Additionally, the end devices can solely update model parameters provided by central servers, thus efficiently sidestepping extensive data transmission and aggregation processes, resulting in improved energy efficiency [50].
- **Cost** aimed at reducing the overall operating cost of services and resources used for the execution of a target application within the edge-to-cloud ecosystem [51].
- **Reliability** in the edge-to-cloud offloading continuum refers to the attainment of successful service delivery, and adhering to its specified requirements and parameters. The primary factors contributing to a decrease in reliability in cloud-to-edge computing environments are the failures or errors that occur in edge devices. This is often influenced by the intermittent connectivity, mobility of the devices or their limited available resources [51].

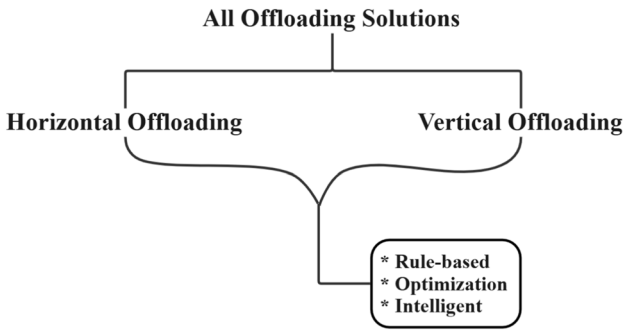


Fig. 5 Classification levels for review

4.4 Evaluation

Evaluation is an important step to guarantee that the offloading mechanisms meet their target objectives. For this purpose, this category focuses on the process of assessing the efficiency and effectiveness of the reviewed offloading solutions. More specifically, we aim to assess the relevant studies in the following aspects:

- **Environment** refers to the experimental environment used by the authors of the candidate solution for the evaluation of their proposed offloading mechanism. The experimentation can be either performed using some simulation tools, e.g., iFogSim, EdgeCloudSim, etc., or the use of a realistic edge-to-cloud setup.
- **Metrics** will list the quantitive indicators used to assess the effectiveness of the proposed offloading solutions in regard to their target objectives. The metrics play a vital role in the decision-making process. This attribute will help in demonstrating what metrics are considered by different solutions to achieve their target objectives such as performance, energy, that were earlier discussed in Sect. 4.3.
- **Workload** refers to the suitability of a given offloading mechanism to the nature of the workload. For example, offloading solutions can be specifically proposed to target generic internet-based systems like [52] or specialized environments such as smart cities [53].

5 Review of existing decentralised computation offloading solutions

This section presents a comprehensive review of existing Decentralised computation offloading solutions in light of the proposed taxonomy. The available landscape of relevant solutions is diverse. Therefore, to reduce the overall complexity and to carry out a useful review, we divided the considered solutions into two levels of classification (see Fig. 5).

The first level based on the scope of an offloading solution categorises it as either Horizontal or Vertical. The horizontal involves two offloading entities within the same layer, e.g. edge-to-edge and fog-to-fog, whereas the vertical offloading refers to those solutions where the involved entities in offloading reside across different layers of the edge-to-cloud continuum. The 2nd level is based on the underlying implementation techniques, earlier explained in Sect. 4, including Rule-based, Optimised, and Intelligent approaches.

This two-level classification, on the one hand, allowed us to cover representative solutions from each category. On the other hand, this enabled us to perform a detailed comparative analysis of existing solutions that are closely related to each other, helping us to cluster the results of each group. The key acronyms are listed in Table 2.

5.1 Horizontal offloading

5.1.1 Rule-based

Qin et al. [54] proposed a distributed threshold-based offloading strategy within the context of large-scale mobile cloud computing, where every mobile (IoT) device decides upon whether a newly arrived task will be processed locally or be offloaded to a cloud node. They followed a game-theoretic approach, where they formulated the problem as a symmetric game. In their approach, the offloading decision-making mainly depends on the value of the threshold that is dynamically calculated and iteratively updated based on the processing delays at the local mobile device and the overall server utilization—calculated using the ratio of the average offloading rate by all devices to the cloud servers' service rate—at the cloud level. The notion of distributed mechanism is from the aspect that the decision-making making happens at the mobile device level, where each one aims to optimise their threshold value without knowing the threshold values of other mobile nodes. Every mobile node updates its offloading decision into a look-up table asynchronously that is used to calculate the overall server utilisation. The communication pattern and protocol between distributed nodes and the look-up table, however, are not clear.

The same strategy is further updated by Qin et al. [55], where they applied it within the context of large-scale mobile edge computing scenarios, however, with the same objective, i.e., each device updates its threshold without knowing all other devices' thresholds. The updated work considered a wide range of metrics for decision-making including the computational load, average offloading latency, average energy consumption at the local device level and edge server processing time indicating server utilization at the edge level. Each device receives the server utilisation through a publish-subscribe mechanism, which can be utilised during the threshold update iteration. Both research works [54, 55] operate in a reactive nature based on the threshold value of decision-making that is dynamically updated iteratively.

Mittal et al. [56] suggested a distributed threshold-based task orchestration framework (DTOF), which supports a vehicle-to-vehicle-based task orchestration and uses the motions of vehicles on metropolitan highways to create vehicular edges. They used a novel and lightweight edge initialisation model in which edge vehicles can seamlessly join and exit the edges. In their approach, by monitoring the STATE of the vehicular edge computing (VEC) server, they developed a method for task orchestration in the proposed vehicular edge that makes distributed decisions about the distribution of communication and computing resources. After completing its task, an edge node vehicle updates the cloud, sends data to the primary server, and leaves. The work is stopped, other vehicles go to IDLE, and the edge reorganises at the next intersection if the main edge vehicle leaves. The communication pattern between distributed nodes and update table is V2V communication, however, the protocol is not clear.

Table 2 List of key acronyms

Acronym	Definition
NE	Nash Equilibrium
HT	Hashing Technique
GT	Game Theory
GA	Genetic Algorithm
P2P	Peer-to-peer
SDN	Software-defined networking
OA	Optimisation Algorithm
DRL	Deep reinforcement learning
FD	Orthogonal frequency-division multiple access
MIL	Mixed-integer linear programming
BC	Blockchain
LTS	Long-term stochastic optimisation
MM	Markov model
FL	Federated learning
DNN	Deep neural network
CNN	Convolutional neural network
AID	Artificial Intelligence-Defined
SE	Stackelberg Equilibrium
FLA	Fuzzy logic algorithm
HA	Heuristic algorithm
RL	Reinforcement learning
RTA	Regression tree Algorithm
SI	Simulation
RW	Real-world
NS	Network Simulator 3
IFS	iFogSim
PY	Python
AL	AnyLogic 8 PLE
MT	MATLAB
CSP	CloudSim Plus
HSP	Hadoop and Spark platform
MNW	Mininet-WiFi
OS	OpenStack
OMN	OMNeT ++
SUM	Simulation of Urban MObility
PES	PureEdgeSim
TF	TensorFlow
ECS	EdgeCloudSim
MIP	Mixed Integer Programming

Yang et al. [57] and Zhou et al. [58] also adopted a distributed threshold-based structure. In the context of combining game-theoretical analysis with constrained nonlinear optimisation theory, Zhou et al. [58] proposed an optimizing offloading decision threshold of agents in a distributed manner. In this case, a device-to-device (D2D) based IoT system is considered for each source agent to offload its computational task to the closest destination or just execute it locally. They employed a mixed-strategy Nash equilibrium. Their method divides the common range that can be used for computation offloading into several frequency channels, while the time horizon-which determines whether to offload the computing work to its destination-is discretized into a series of times. The concept of a distributed mechanism stems from the perspective that a computation task is divided and assigned to multiple devices or nodes in a parallel and separate manner with cooperation. The source agents determine their optimal thresholds for offloading decision-making in parallel using an embedded, unconstrained Lagrangian optimisation (ULO) scheme. Although P2P is the pattern used for communication between dispersed nodes, which guarantees effective and decentralized communication.

Yang et al. [57], however, presented a potential game-based offloading algorithm (PGOA) to reach a Nash Equilibrium (NE), in which mobile devices (MDs) decide whether to offload computing at every choice. They follow a possible game-based offloading technique that enhances the cloud computing capabilities of mobile devices (MDs), which allows MDs to change their choice after each iteration while minimising overhead. According to their method, all offloading choices by mobile devices reach the NE state, meaning the best solution and better response rules for the overhead minimization problem. For decision-making, several metrics are considered in this work including latency, execution delay, local computer overhead, energy consumption, data transmission rate, and energy consumption of computing jobs. Although Zhou et al. [58] demonstrated diverse IoT devices equipped with D2D communications, Yang et al. [57] investigated a distributed computation offloading strategy for a system involving multiple devices and servers in the context of orthogonal frequency-division multiple access. While Zhou et al. [58] mostly focus on energy and delay, Yang et al. [57] primarily concentrate on the offloading rate.

Chen et al. [59] proposed a distributed horizontal resource allocation strategy developed by assigning different tasks among two-layer clouds within the context of vertical and horizontal cloud federation networks. In these networks, users can process excessive requests from other devices by sharing duplicate resources on each device. The authors follow a game-theoretic model that considers both cooperation and competition among cloud providers in the joint vertical and horizontal cloud federation (VHCF) Provide full form. network. In their approach, resource allocation primarily depends on the coalition value, which is determined by the workload factoring game. what does the coalition value represent The coalition value reflects the overall benefit that a group of private clouds can achieve by collaborating, taking into account both the resources they share and the workloads they handle. The specific coalition structure also influences this value, meaning that the benefits depend not just on the coalition itself but on how the clouds are grouped. To manage this complexity, the authors proposed two algorithms for different levels of the problem. At the lower level, they suggested CLOSE and CLONE–the latter being a generic algorithm utilizing the constraint generation method–to handle the intricate nature of coalition value functions. At the upper level, they proposed Brute-Force and CUBE, which use a binary search strategy guided by two heuristic principles to find the optimal coalition structure. These algorithms aim to find the most efficient and stable cooperation patterns among clouds. Through horizontal federation, the communication pattern and protocol adhere to a client-server architecture based on a gossip protocol, facilitating the pooling of resources between private clouds. This may improve the level of service quality. They mostly focus on delay and runtime management as a cost. In addition to theoretical approaches, Chen et al. [60] also consider practical implementations by investigating a hierarchical EETORP (Economic and Efficient Task Offloading and Resource Purchasing) framework that incorporates a two-stage joint optimisation procedure. The problem's NP-completeness was then demonstrated. In order to demonstrate the existence of at least one Nash equilibrium strategy in MCA-Game, they first defined the offloading problem as a multi-channel access game (MCA-Game). They then examined the performance guarantee of the resulting offloading technique in the worst-case scenario and suggested a game-based multi-channel access (GMCA) approach to find the Nash equilibrium strategy. In the second step, they employed Stackelberg game theory to describe the distribution of computing resources between users and edge servers (ESs) in the edge layer. The goal of the proposed EETORP is to minimize the cost in the context of real locations of base stations and UDs in metropolitan Melbourne, Australia.

Li et al. [61] proposed a distributed implementation of an energy threshold-based offloading strategy that analyses the time and energy consumption involved in service execution and data transmission. They follow an algorithm that converges to the Nash Equilibrium offloading decision by calculating the energy consumption and latency cost of task execution for local terminals, edge servers, and the central cloud-to construct a computation offloading model. The notion of a distributed mechanism arises from the fact

that each service can be divided into multiple independent and tandemly executed sub-services, which can migrate from the mobile server to the edge server corresponding to this type of sub-service for computing. Every mobile node updates its offloading decision based on the threshold-based service migration strategy to solve the network congestion problem caused by the unbalanced load of edge servers and the mobility of users. The communication pattern between distributed edge nodes and end devices follows server/client patterns. Similar to this, Tang et al. [62] also proposed a decentralized threshold-based approach for mobile edge computing in which there are geo-distributed mobile edge servers that are dispersed geographically. In their approach, mobile devices can easily obtain the available information of interested mobile edge servers from the nearest network-connected mobile edge server-managed by a peer-to-peer (P2P)-enabled decentralized mobile edge server management scheme-to improve offloading decisions for selecting the target mobile edge server. All mobile edge servers can participate in updating information in a decentralized manner. The communication pattern between distributed edge nodes is P2P. Both studies [61, 62] followed a reactive approach, where they iteratively updated threshold values for decision-making.

Al-Hammadi et al. [63] presented a distributed threshold-based offloading technique within the context of software-defined networking (SDN), providing a worldwide perspective of the network. In this framework, task requests are collected and allocated to suitable Mobile Edge Computing (MEC) servers capable of meeting the demands. The authors employed four scheduling algorithms to address challenges such as overloaded tasks, congested networks, and emergency offloading. In their approaches, offloading decision-making depends on various factors like the distance of the MEC server, calculated using its location and available resources to avoid wastage; the threshold on the MEC server to alleviate network congestion; priority of emergency tasks, using the Scheduling Emergency Task Algorithm (SSETS); and tasks' deadlines, using the Stealing Regular Task for Emergency Tasks Scheduling Algorithm (SRETS). These algorithms are controlled by a framework consisting of three planes—the control plane, the data plane, and the infrastructure plane—collaborating to provide MEC servers with the processing power and low latency they need for long-running applications. The communication pattern between distributed nodes is hierarchical. This study represents an extension of Phan et al. [64], who implemented a dynamic fog-to-fog offloading solution for SDN-based fog computing systems. The SDN controller determines whether to process a task locally or on the edge cloud based on the availability and load of various edge clouds. Numerous metrics such as computational resources (e.g. CPU, memory, and storage) and network resources (e.g. bandwidth, communication hop, and latency) are considered for decision-making in this work. Both studies followed a reactive approach to decision-making.

Sulimani et al. [65] proposed a Hybrid– both dynamic and static algorithms– decentralised cluster-based energy-efficient task offloading strategy in the realm of fog computing, where individual clusters comprise a solitary fog node (FN) along with multiple terminal nodes (TNs) that delegate their tasks to the FN within their respective clusters. Their methodology was based on a fog architecture utilizing Software-Defined Networking (SDN), wherein a dual-objective problem was formulated to achieve the optimal clustering of TNs to FNs, selection of offloading parameters, and the joint minimization of delay and energy consumption. They implemented an optimal clustering and Offloading scheme (OCOP), which prioritises energy efficiency while considering the delay constraints for task offloading among nodes in the fog layer. The communication pattern is client and server/and the protocol between distributed nodes is zab.

Trust management and security in edge-to-cloud continuum utilities advanced techniques (e.g. blockchain and game theory), which combine rule-based approaches with intelligent, decentralised models. Esposito et al. [66] propose a blockchain-based trust management system in the context of edge computing, using Bayesian Signaling Games (BSG) and Dempster-Shafer theory in order to detect and reject fake trust scores from IoT nodes. In their approach, a game-theoretic approach is employed to prevent deceptive interactions with scalability and energy efficiency. In their combination, the signalling game helps to detect suspicious nodes at the interaction level and the evolutionary Dempster-Shafer removes wired outliers to combine multi-reputation scores. The notion of distributed mechanism is from the dynamically implemented and incorporated of a Hyperledger-based blockchain platform that protects the integrity of decentralised trust management in the face of adversarial conditions. The proposed approach is adaptive and context-aware, leading to effective protection against false data injection in the edge-to-cloud continuum. In terms of results, a testbed scenario based on the gossip protocol and the star topology for the sensor network is implemented. Similarly, Chen et al. [67] present a game-theoretic task offloading technique for multi-source data in mobile edge computing (MEC). They follow a formulation problem as a distributed optimisation game, allowing each edge node to make independent decisions based on local conditions. In their approach, a Nash equilibrium in a distributed system is employed for load and task completion time balancing, solving challenges in heterogeneous contexts, such as data privacy, high complexity, and load imbalance, while demonstrating lower computational overhead

and better utilisation compared to centralised approaches. Although both approaches rely on decentralised game-theoretic techniques, Esposito et al. consider secure trust evaluation in adversarial scenarios, whereas Li et al. focus on performance optimisation on distribution efficiency for edge computing. The communication pattern between distributed nodes is P2P communication. Esposito et al. employ gossip protocol, however, the protocol used with Chen is not clear.

Table 3 summarizes the horizontal rule-based offloading discussed in this subsection. The table is not exhaustive, as our purpose is to identify the weaknesses of the most developed works. Most of the identified limits of horizontal rule-based solutions can be overcome by P2P architecture (e.g., [56] and [58]). Most research papers cover the performance and energy domains. Game theory is the most commonly utilised approach in rule-based offloading (e.g., [57] and [59]). Future rule-based offloading solutions may use proactive operation type to address the pending issues (e.g., orchestration and real-time applications).

5.1.2 Optimised

Chen et al. [68] proposed approach leverages mobile edge computing (MEC) to effectively reduce delay in UAV networks by offloading computation tasks to the master node. They employed a scheduling method based on the shortest effective job first (SEJF) approach, which addresses the interplay between scheduling and resource allocation to solve computation offloading and channel access challenges. Their approach relies on the concurrent best-better response (CBBR), which enables the selection of independent members whose decisions do not impact the utility of others. Their approach is characterised by edge-level decision-making, where each participant aims to optimise resources in distributed wireless networks due to its computational simplicity and effectiveness. The offloading data is dynamic since it depends on more than just the UAV's hardware. The communication pattern is public/subscribe since UAV members have distinct areas of responsibility. Reliable awareness and delay awareness are supported in the proposed offloading technique. Mensah et al. [69] put forth a distributed game-theoretic approach to vehicular edge computing, where they applied it within the context of device-to-device communication, however, with a connection between vehicles and coordinates with the edge and cloud to allow for flexible allocation of computing resources among vehicles and mobile users. Their approach involved the creation of a software-defined network (SDN) within the vehicular network (SDVN). This network was situated on a unidirectional road without obstacles and had uninterrupted traffic the vehicles were assumed to be moving at a constant speed along the road. The concept of an SDVN is inferred from the idea that all pooled edge resources are not allocated to a single entity or a specific type of computing that determines the proportion of edge computing resource allocations for VEC and MEC. The communication pattern between distributed edge nodes is peer-to-peer (P2P). This research is further enhanced in the study conducted by Fang et al. [70], where they examined the application of the D2D-assisted heterogeneous user cooperation framework, which allows two users with varying demands to share their idle local resources, all the while pursuing the same objectives and purpose. To achieve this, all users broadcast their local resources and request assistance from their neighbouring users. To determine the stable matching in this proposed game theory, the authors modify the traditional deferred acceptance (DA) algorithm and introduce the multi-round cooperation matching (MRCM) algorithm that consists of two main components like D2D matching, in which both compute-intensive users (CIUs) and content-desired users (CDUs) construct their preference lists based on the information they have collected, and edge matching. CIUs and CDUs communicate their demands and available local resources using a publish/subscribe protocol. This is achieved by broadcasting over a shared control channel, allowing all participants to efficiently share and receive relevant information. The communication pattern between the distributed edge nodes is peer-to-peer (P2P). All of the aforementioned studies [68–70] followed a reactive approach to decision-making.

Wakgra et al. [71] proposed a distributed deterministic policy gradient (DTD3) approach, where a multi-objective offloading optimisation in multi-access edge computing (MEC) and vehicular-fog (VF) systems. The strategic position is closer to Internet of Things (IoT) devices, reducing propagation latency compared to cloud-based solutions and ensuring satisfactory quality of service (QoS). They follow a deep reinforcement learning (DRL)–generally reactive–agent, the dynamic distributed twin-delayed deep deterministic policy gradient (DTD3) algorithm, deployed at the control plane in order to process incoming traffic locally or offloaded to a nearby horizontally federated MEC site or a vertically federated VF to enhance response times. The algorithm's main objective was to make an optimal offloading decision that simultaneously minimizes latency and maximizes battery life for the VFs while satisfying QoS requirements. The communication pattern between the distributed nodes and the updating table follows a peer-to-peer model, the communication pattern, however, is unclear.

Bi et al. [72] approach focused on mobile-edge computing, which involves scheduling tasks from resource-limited smart mobile devices (SMDs) to high-performance edge servers. They follow a hybrid metaheuristic algorithm like genetic simulated annealing-based particle swarm

Table 3 A summary of the literature on rule-based horizontal offloading

Category	Reference	Architecture									Strategy					Objectives				Applied approaches	Evaluation tools
		Pattern				Protocol					Resource allocation		Operating type								
		P2P	Hierarchical	Publish/sub	Client/ server	Gossip	Blockchain	Paxos	Zab	Raft	Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Rule-based	[54]											✓	✓			✓				NE	SI
	[55]											✓	✓			✓	✓			NE	SI RW
	[56]	✓														✓		✓		HT	NS3
	[57]	✓														✓	✓			GT	SI
	[58]	✓														✓				GT	SI
	[59]				✓	✓					✓		✓			✓		✓		GT	RW
	[60]		✓						✓		✓		✓					✓		GT	RW
	[61]				✓							✓	✓			✓	✓			GA	SI
	[62]	✓										✓	✓			✓	✓			P2P	NS3
	[63]		✓										✓			✓	✓			SDN	RW
	[64]	✓								✓		✓	✓			✓			✓	SDN	RW
	[65]				✓				✓		✓	✓	✓			✓	✓			OA	IFS
	[66]	✓				✓		✓	✓										✓	GT	RW
	[67]	✓									✓		✓			✓			✓	GT	ECS

optimisation (GSP) to formulate the problem as a nonlinear constrained optimisation. In their approach, the decision-making process for offloading is primarily influenced by the multiple serving access points (APs)—equipped with a pool of edge servers—that offer SMDs proximity and flexible access to services provided by remote cloud data centres. The partial mechanism in this approach is characterized by the decision-making occurring at the edge servers that aims to optimise the task offloading ratio, CPU speeds, and transmission power for each SMD individually. The partial edge computing scheduler divides the data of all SMDs into independent parts, enabling parallel execution in which each edge server updates its offloading decision and particle velocities in the particle swarm optimisation (PSO) algorithm. The communication pattern between the distributed nodes and the updating table follows a peer-to-peer model. This study is further enhanced in the work of Bi et al. [73], where they implemented it in the context of energy-efficient task offloading using hybrid particle swarm scenarios, however, their objective remained the same, i.e., edge server updates its offloading decision and the velocities of particles in the particle swarm optimisation (PSO) algorithm without knowing other devices. The updated study considered a wide range of metrics for decision-making, including latency limits, CPU computational speeds, uplink/downlink transmission power limits, the maximum energy of small mobile devices (SMDs), and the number of CPU and memory resources in edge servers. The scheduling of edge servers is obtained by connecting to a serving access point (AP) that possesses multiple servers and offers flexible and nearby services to SMDs. Both of these studies function reactively based on the dynamically updated PSO value of decision-making.

Ko et al. [74] approach offload tasks in a way, where each mobile device maintains dedicated task buffers and tasks are executed sequentially. To determine whether a task should be offloaded, the mobile device—referred to as the task owner—checks its current resource availability, if the resource is heavily occupied by other applications, making it impossible to complete the task by the deadline, the task is offloaded to other mobile devices. The offloading decision is primarily based on four tuples including energy consumption, input data, output data, and deadline that are broadcasted by the task owner–a task owner opportunistically broadcasts an offloading request that includes its mobility level and task completion deadline–during the offloading request. In this approach, nearby mobile devices use their background load and information from the offloading request message of the task owner to determine whether to accept the request or not, based on their policies. The policies are specific to each mobile device. Therefore, multiple mobile devices can accept the same request, which can lead to Energy inefficiency from a global perspective. To address this issue, mobile devices exchange their temporary policies using time-slotted WiFi directly and adjust their policies based on those of other mobile devices. Lee et al. [75], however, introduced a distributed task offloading algorithm, known as neighbour-aware distributed task offloading algorithm (NA-DTOA), wherein the IoT device considers its energy status and the decisions of neighbouring devices when determining whether to process the task autonomously or offload it to a selected edge. They followed a constrained stochastic game model, wherein the policies of IoT devices are updated iteratively using a best response strategy based on Nash equilibrium, to enable distributed implementation of the offloading service. The decision-making process in their approach is primarily influenced by the local state space, local action space, transition probability, cost function, and constraint function within the game model, thereby facilitating the development of an energy-aware offloading system. Both these works operate in a reactive nature based on the energy status of devices. The communication pattern of both studies between distributed devices is D2D with a dynamics-based response algorithm. The distributed system is reliably and systematically disseminated by utilising the concepts of the Zookeeper Atomic Broadcast (ZAB) protocol, as emphasised in [74], where the task owner broadcasts its offloading request, including identification, mobility level, and task deadline, and in [75], where IoT devices broadcast their offloading policies while keeping open receive windows to collect responses.

Saleem et al. [76] considered a network with a smart base station (SBS) and multiple user devices, where users have the option to offload their computations either through device-to-device (D2D) communication or through SBS offloading. They utilised an integrated approach that incorporates partial offloading and interference management using an orthogonal frequency-division multiple access (OFDMA) scheme. The problem formulation aims at minimising total latency as a mixed integer nonlinear programming (MINLP) problem by considering constraints related to energy consumption, partial offloading, and resource allocation. They introduced a novel scheme called joint partial offloading and resource allocation (JPORA), which adjusts the data segmentation and solves the QoS-aware resource allocation problem for cellular links, followed by interference-aware resource allocation for D2D links. The notion of distributed mechanism is from the aspect of the communication and computation resources, where each task device (TD) offloads its entire task for remote execution and user association for the selection of spectrum-sharing pairs independently. The independent offloading scheme is used to determine the fraction of task data to be offloaded at each TD. The communication pattern between distributed

devices is peer-to-peer (P2P). Similarly, a distributed deadlock-free task offloading algorithm was proposed by Zhang et al. [77], where users can integrate communication, sensing, and computing cloud with data-dependent constraints. In their approach, to achieve a shared offloading assignment, they suggested a distributed deadlock-free task offloading (DDFTO) algorithm that ran concurrently on each satellite, switching between sub-task inclusion and consensus and sub-task removal. Their strategy is to prevent deadlocks caused by the addition of subtasks with dynamic insertion positions of sub-tasks based on interval relationships. The idea of distributed originates from the distributed deadlock-free task offloading (DDFTO) method, which uses local communication via ISLs to run on each satellite in parallel with the proactive operating type. The communication pattern and protocol between distributed nodes, however, are unclear.

Iqbal et al. [78] presented a strategy for reputation management within the micro-level vehicular fog networks, focusing on end-users and computing units involved in resource-sharing in terms of delay and energy consumption. They focused on micro-level edge computing units deployed at intersections where a high number of vehicles are available for resource sharing. In their strategy, the decision to offload tasks depends on the deployment of micro-level edge computing units, which provide Quality of Service (QoS) to resource-limited end-users (vehicles) by using computing resources within the range of data transmission. Based on the vehicle's current workload and reputation in society, the vehicle selection model is reactive. The concept of a distributed mechanism is incorporated into their framework, allowing vehicles with onboard computing support to execute tasks dynamically like locally or outsource them to other devices with enough roadside units (RSUs), which receive offloaded tasks from vehicles and assign them to nearby fog vehicles. The overall reputation management framework is based on blockchain technology, with RSUs deployed at intersections and moving vehicles acting as fog nodes. The communication pattern between distributed nodes is peer-to-peer (P2P), while the protocol is based on blockchain technology. Similarly, Beak et al. [79] implemented a privacy-preserving and reliable D2D offloading scheme in the context of mobile device scenarios, however, using the same protocol and pattern, each device also can offload tasks to neighbouring mobile devices using wireless communication technologies such as WiFi-Direct and Bluetooth. The nature of the task is dynamically determined by the size of the data and the amount of computing resources required, specifically the number of CPU cycles. The task owner transfers this information through D2D communication. Both of these studies operate reactively based on the blockchain protocol and D2D pattern.

Yue et al. [80] proposed a hierarchical strategy for distributing tasks and resource allocation. Their approach proposed user devices with different capabilities are allowed to offload their computational tasks to edge servers. They developed online task offloading with delay guarantees (TODG)—an algorithm to solve the long-term stochastic optimisation problem. In this paper, they used the Raft consensus protocol to guarantee uniformity and coordination between user devices and distributed edge servers, allowing the suggested TODG online algorithm for long-term stochastic optimisation to be implemented efficiently in parallel. This algorithm is implemented in parallel among user devices and edge servers to ensure worst-case delay guarantees for all offloading tasks. They also introduced a dynamic strategy for making offloading decisions to reduce the computational cost and communication overhead associated with the complex computation required for resource allocation at the edge level. The distributed mechanism in this approach is designed to handle diverse computational tasks with varying attributes, such as size, resource requirements, and response times, which need to be offloaded to a selection of edge servers with different computing capacities, using a limited number of random communication channels. Each edge server updates the task queue and delays the state queue to make optimal channel allocation decisions in a distributed manner using the multi-robot assignment algorithm. Avgeris et al. [81] also proposed a distributed resource allocation technique for large-scale edge computing scenarios that considers energy consumption while aiming for low latency. The approach balances the trade-off between minimising total energy consumption, meeting end-to-end delay requirements, and maintaining load balancing at the edge. They followed a performance prediction modelling approach based on the switching systems theory—switching between the different hardware profiles— to have the dynamic selection and allocation of resources for each application. To optimise energy management, they combined this approach with a novel Markov Random Field mechanism, which facilitates efficient state space search in a distributed manner. The notion of a distributed mechanism appears from the need to redirect tasks that cannot be executed locally due to energy and quality of service requirements in which resource utilisation is balanced throughout the entire infrastructure. The communication pattern between distributed edge servers is peer-to-peer the protocol, however, is unclear.

Li et al. [82] and Su et al. [83] proposed distributed multi-cloud offloading strategies considering large-scale cloud computing environments. Li et al. [82] approach consists of an architectural framework for fault-tolerant scheduling and data placement of cloud systems. This framework includes a controller component that calculates

the appropriate bandwidth between datasets and data centres, while also optimizing time and performance across different dataset locations within the data centres. They employed a speculative spark-based execution method on master nodes to determine the load state of the cluster where the data is stored. Their approach involves a hierarchical processing structure of data streams, consisting of two distinct phases: data stream creation execution to minimize bandwidth expenses. The communication pattern between masters and client nodes is hierarchical. Similarly, Su et al. [83] also presented a game-based distributed pricing and task offloading approach in the context of multi-cloud and multi-edge environments. However, despite the similarity in terms of domain and objectives, this approach allows each client device to make offloading decisions to execute locally in the MD or remotely on the edge servers without relying on the master node. They followed a multi-leader multi-follower two-tier Stackelberg game model within the context of the multi-edge and multi-cloud, which considers the dynamic interaction between service providers and mobile devices (MDs). The communication pattern between distributed edge servers is client/server the protocol, however, is unclear. While both these works aim to improve execution delay and energy consumption reactively, Su's work also considers cost.

Hassija et al. [84] introduced a distributed offloading framework that combines blockchain and virtual voting in the context of mobile cloud computing, where the framework enables mobile users to allocate computational tasks to alternative resource-rich mobile devices, thereby reducing energy consumption and improving performance. They followed a game-theoretic approach to model the interactions between mobile devices, recognizing that peer-to-peer connections among these devices can be a highly effective solution for time-sensitive offloading tasks. To address the limitations of generic blockchain technology on mobile devices, they developed the directed acyclic graph-enabled mobile offloading (DAGMO) algorithm that leverages traditional blockchain features and offers additional benefits to overcome these limitations. The notion of distributed mechanism in this approach occurs at the mobile device level, where a secure distributed ledger technology (DLT)-based approach creates a peer-to-peer network of mobile users for offloading data and computation tasks. Mobile devices with high computation power continuously update their computation time values until they reach their threshold values in which a high computation power device only updates its computation time if it was not the winner in the previous iteration, ensuring its chances of winning the present iteration. Jing et al. [85] introduced resource trading and computation offloading in the domain of mobile edge computing scenarios, however, with the same technology and workplace, i.e., both resource trading and computation offloading strategies are considered. They followed an incentive mechanism within a blockchain-based resource-sharing system, which encourages more users to engage in resource sharing. Their approach utilizes a one-leader and multi-followers Stackelberg Game for the resource trading process, aiming to maximize the utilities of both requesters and providers. This study incorporates various metrics for decision-making, such as local CPU frequency, transmission time, and transmission bandwidth to energy consumption. Both of these studies exhibit a reactive nature, operating on a peer-to-peer connection pattern, with the communication protocol between distributed nodes being blockchain-based.

In Hu et al. [86] approach, each mobile user decides to offload tasks to a MEC server or a distributed computing node (DCN). They followed a multi-user and multi-destination computation offloading scheme that aimed to achieve the Nash equilibrium. By considering the set of mobile users as a finite set of players, they established the decision-making problem for multi-destination computation offloading as a sequential game, wherein each mobile user made decisions based on perfect knowledge. This research work considered several metrics, including channel bandwidth, transmit power, D2D distance, and receive power consumption. Both studies followed a reactive approach to decision-making.

While the above studies are mostly theoretical, some works have also explored real-world implementations, such as Zeinab et al. [87] and Tian et al. [88]. Zeinab proposed a testbed optimisation: Smart Mobility Services On The Edge Computing (SMOTEC) for adaptive smart mobility experimentation using edge computing. They employed a flexible and extensible Application Programming Interface (API) that integrated networked Raspberry Pi devices and EPOS's multi-agent collective learning approach on edge-to-cloud computing infrastructures. In their approach, all modules can be easily replaced with others to fulfil the needs of various applications and experimental scenarios, leading to load balancing and optimal service placements. In terms of implementation, they utilised a lightweight Kubernetes distribution (k3s) with little memory, simple configuration, and low resource consumption–suitable for IoT devices–that is quick to set up and fast to auto-update. As a result, a distributed real-world load-balancing scenario is demonstrated for service placement in the context of monitoring traffic flows in Munich. Tian et al. [88] also proposed an Indoor Connected Autonomous Testbed (ICAT) for vehicle computing and V2X connectivity at the edge computing. They follow a cost-effective and scalable solution that focuses on V2X communication and vehicle computing, allowing for reliable data exchange between vehicles, infrastructure, and cloud servers. Their approach overcomes the weaknesses

of other indoor self-driving systems, which frequently lack enough sensing and computation capabilities. They employ digital twin technologies–including simulators (such as SUMO, CARLA, and LGSVL) and a custom Python-based multi-agent simulator–to enable high-level development and testing of centralised and decentralised autonomous driving tasks. Therefore, their approach grants comprehensive research in multi-agent navigation, swarm intelligence, and real-time traffic optimisation.

There are several works presented in this subsection, indicating that research on horizontal optimised offloading approaches is still needed more study. Many of the optimised solutions described in Sect. 5.1.2 claim a form of P2P offloading. However, some other works are using other patters (e.g., [80] and [82]). Table 4 provides a summary of the optimised solutions discussed in this subsection. The main findings are that no existing work has used gossip and paxos protocols, and raft and zab still need further development. In summary, future research should include the horizontal offloading system across all protocol domains (e.g., gossip, paxos, and raft) to achieve the most effective and efficiently optimised offloading framework, regardless of the specific operation type being targeted.

5.1.3 Intelligent

Gue et al. [89] introduced a distributed machine learning strategy within the context of multiuser mobile edge computing (MEC) networks, where several users possess computational tasks of varying priorities. These tasks can be offloaded partially or fully to multiple computational access points (CAPs) for processing. They adopted a distributed machine learning approach to optimise the offloading ratio, bandwidth allocation, and computational capacity while prioritising tasks. Their objective consists of three key criteria including, minimisation of latency and energy consumption overall, minimisation of latency while maintaining a specified energy consumption threshold, and minimisation of energy consumption under a set latency threshold. The concept of distributed mechanism is rooted in the idea that users exhibit diverse task characteristics, such as task length and priority, necessitating dynamic allocation of communication and computational resources to boost system performance by reducing latency and energy consumption. The computation is performed simultaneously by multiple users, with each user utilizing their own computational resources-either local or provided by the Cloud Access Providers (CAPs)-to process the tasks. The communication model between distributed nodes is peer-to-peer. Hu et al. [90] also proposed an energy-focused distributed offloading model, leveraging device-to-device communication using a federated learning approach. This work focused on addressing the impact of varying energy levels among devices and the fluctuating energy levels of individual devices. In their approach, they introduced a federated edge learning (FEL) system with Zab protocol–it will broadcast the model to all devices for the next round of training–, where a base station (BS) orchestrates a group of edge devices to train a shared machine learning model cooperatively. They followed a stochastic optimisation problem and devised an online control algorithm based on Lyapunov optimisation to determine data offloading, queuing, and processing requirements for each device at every time slot during training. Both these works exhibit a reactive approach, guided by the P2P communication framework of decision-making that is dynamically updated iteratively.

Tadele et al. [91]investigated how information transmission and response times affect end-to-end AoI and developed an edge-enabled vehicular fog system to reduce the computational load on IoT devices. In order to compare its performance with DQN policy and analytical results, they created a deep reinforcement learning (DRL)-based algorithm called Dueling-Deep Queue Network (dueling-DQN). In their approach, they have incorporated DRL-based agents into their strategy to enable rapid decision-making and interaction with the surroundings. For the purpose of optimising AoI in the intended environment, they specified the agent, states, actions, and incentives. They follow an edge-enabled vehicle fog environment that gives the agent information as a state to decide what to offload. The offloading agent then used the existing network to update the target network Q-value as well as the current Q-value. Based on the gradient loss values, this updating procedure is carried out. The concept of a distributed mechanism is based on the idea that in edge-enabled vehicular fog environments, offloading decisions must be made at each point in time in order to achieve their goals. This study exhibits a proactive nature, operating on a client/server connection pattern, with the communication protocol between distributed nodes being Raft.

Chen et al. [92] introduced an unsupervised deep learning method based on the DNN model for binary offloading within the framework of multiuser mobile-edge computation offloading (MECO), where tasks are either processed locally or transferred entirely to the edge servers. They employed Time Division Multiple Access (TDMA) to reduce the energy consumption of mobile devices (MDs). Their strategy revolved around making offloading decisions based on the multiuser binary MECO model operating within a TDMA system, formulated as a mixed integer programming (MIP) challenge and the optimisation strategies —including offloading decisions, transmission time slots, and transmission power—are collectively fine-tuned to minimise the total energy consumption of MDs. The

Table 4 A summary of the literature on optimised horizontal offloading

Category	Reference	Architecture									Strategy					Objectives				Applied approaches	Evaluation tools
		Pattern				Protocol					Resource allocation		Operating type								
		P2P	Hierarchical	Publish/sub	Client/server	Gossip	Blockchain	Paxos	Zab	Raft	Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Optimised	[68]			✓								✓	✓			✓			✓	NE	SI
	[69]	✓											✓					✓		GT	SI
	[70]	✓		✓					✓				✓			✓				GT	SI
	[71]	✓										✓	✓			✓	✓	✓		DRL	PY
	[72]	✓										✓	✓			✓	✓		✓	GA	RW
	[73]											✓	✓			✓	✓			GA	SI
	[74]	✓							✓			✓	✓				✓			NE	SI
	[75]	✓							✓			✓	✓			✓	✓			GT	SI
	[76]	✓														✓	✓			FD	SI
	[77]											✓		✓						MIL	PY
	[78]	✓					✓					✓	✓			✓	✓			BC	AL
	[79]	✓					✓					✓				✓			✓	BC	SI
	[80]		✓							✓		✓	✓			✓		✓		LST	MT
	[81]	✓										✓	✓			✓	✓			MM	CSP
	[82]		✓									✓	✓			✓	✓		✓	OA	HSP
	[83]				✓							✓				✓	✓	✓		GT	SI
	[84]	✓					✓				✓		✓			✓	✓			BC GT	MT
	[85]	✓					✓						✓			✓	✓			GT	SI
	[86]	✓										✓	✓			✓		✓		NE	SI
	[87]	✓										✓						✓		OA	RW

concept of a distributed mechanism is rooted in the idea that the CPU of terminal MDs can handle the computation-heavy tasks locally or remotely delegate them to other edge servers through offloading, all under the TDMA framework.

Huang et al. [93] followed a distributed supervise deep learning-based offloading approach within the framework of mobile edge computing, where each wireless device (WD) makes decisions regarding offloading computation tasks to the edge server. They utilised multiple parallel deep neural networks (DNNs) to efficiently and effectively generate these offloading decisions. This algorithm for distributed deep learning-based offloading relies on solving bandwidth allocation issues, such as transmission power allocation at WDs and computing resource allocation at the edge server, which can be substituted with other dedicated components in the algorithm structure. Their approach utilises multiple parallel deep neural networks (DNNs) to generate offloading decisions efficiently and effectively. These decisions are saved in a shared memory for further training and enhancement of the DNNs. The communication pattern between distributed edge servers is client/server the protocol, however, is Raft which is based on server nodes.

Li et al. [94] introduced a distributed deep reinforcement learning algorithm intended to address the intricate computation offloading challenge within the domain of collaborative computing in heterogeneous Edge Computing Servers (ECS), where each edge server responsible for part of the task to effectively use edge network resources. They utilised artificial intelligence technology, driven by the emergence of deep learning, to offer innovative solutions to complex model issues. Their methodology entails selecting edge servers across diverse edge subnets to tackle the multi-user edge computing offloading predicament through a distributed computing approach, utilising Deep Deterministic Policy Gradient (DDPG) to allocate various task types to suitable locations within the edge subnet for processing. The concept of a distributed mechanism stems from the perspective of autonomous edge subnets' agents, where each one aims to access other subnets' network state via a distributed file system to determine computation and offloading actions based on task requirements. The communication pattern among distributed nodes is publish/subscribe, the protocol, however, remains ambiguous. Similarly, Naveen et al. [95] introduced an edge-to-device edge computing framework aimed at optimising the deployment of models within the context of small smart devices designed for time-sensitive applications through weight pruning. Their approach involved a comprehensive distributed deep (DL) framework—utilising a pre-trained DL model from the cloud—that coordinates edge clusters to handle diverse local data to run real-time applications with minimal latency. They developed a collaborative horizontal convolutional neural network (CNN) on IoT devices with limited resources to enhance their intelligence, where input feature maps are divided and distributed among resource-constrained edge devices to reduce memory usage and latency. This approach involves loading the optimised pre-trained CNN model like Yolov2 onto edge clusters with multiple edge nodes, where each edge node makes task decisions based on the model. Subsequently, the gateway device integrates results from all edge devices to conduct collective inference and present the total latency for the inference task. Both studies exhibit a reactive approach based on reinforcement learning that is continuously updated iteratively. The communication pattern among distributed nodes is hierarchical, the protocol, however, is unclear.

Qiu et al. [96] introduced a distributed and collective Deep Reinforcement Learning (DRL) based technique within the context of Mobile Edge Computing (MEC), where each mobile device kept the capability to dynamically make decisions regarding offloading and channel allocation without having full knowledge of the system model. They formulated the optimisation issue as a Markov Decision Process (MDP) and utilised recent advancements in DRL to cater to the fluctuating user demands and account for long-term efficiency. In their strategy, the offloading choices are determined based on the system's state including aspects like the tasks generated, the edge server (ES), a group of mobile users (MUs), and the wireless channels. They utilised a distributed training system, where multiple MEC environments collectively train DRL agents and update network parameters. Each mobile user updates parameters utilising the prediction policy gradient obtained from the backpropagation of the value function following the deterministic policy gradient theorem. The communication pattern between distributed edge nodes is hierarchical the protocol, however, is Raft which is based on edge servers.

Yang et al. [97] presented a two-layer distributed online task scheduling framework within the context of optimising the Task Acceptance Ratio(TAR) while addressing unbalanced task distribution and diverse Quality of Service (QoS) requirements. Their approach involved implementing a dynamic priority-based transmission scheduling policy specifically designed for vehicles. In their approach, the decision-making process for offloading tasks primarily relies on a distributed task dispatching policy, where each edge server—connected based on P2P communication—can autonomously handle task offloading requests from vehicles. In This approach, each edge server is responsible for making task assignment decisions and dispatching tasks to process servers that subsequently execute tasks and deliver the results back to the vehicles. The communication pattern between distributed edge nodes is P2P the protocol, however, is Raft.

Shah et al. [98] presented a distributed artificial intelligence-defined wireless networking (AIDWN) for computational offloading and resource allocation by extending the traditional Software-Defined Networking (SDN) within the context of edge computing layer. They followed a deep reinforcement learning framework and deep neural networks to dynamically adjust optimal computational offloading and wireless resource allocation decisions considering challenges such as handover, mobility, and coordinated resource allocation in dynamic and mobile multi-MEC server environments. The approach involved intelligent offloading decision-making capabilities to enhance data flow management by analysing real-time wireless network data, application-specific requirements, and MEC server capabilities to predict time-varying wireless channel conditions. The concept of a distributed mechanism comes from the agent's ability–reinforcement learning agent acts on the SDN controller–to learn the optimal target policy through either online or offline training. The Real-time execution of offloading decisions and resource allocation configurations was achieved by the SDN controller responsible for deploying dynamic flows, routing, and control updates in the data plane devices. The communication protocol is not clear.

Shi et al. [99] approach deals in real-time pricing mechanisms utilising deep reinforcement learning considering vehicular computing, where a vehicle can delegate a portion of its computational tasks to nearby vehicles within its communication proximity. They utilised the Deep Q-Network technique to effectively distribute all tasks to the available computing resources within a highly dynamic vehicular setting. In their approach, the offloading decision relies on the condition of wireless channels and the duration of links between the task vehicle (TV) and service vehicles (SVs). The base station (BS) servers, representing the edge layer, work as the agents overseeing the overall task distribution. The mode of communication between distributed vehicles and the Q-value table is V2V, the protocol is Raft.

Intelligent horizontal offloading in edge-to-cloud solutions, as well as their inherent limitations, were discussed in this subsection. The P2P pattern is still the most used architecture for horizontal offloading (e.g., [97] and [99]). In Table 5, intelligence offloading approaches are compared for various architectures, strategies, and objectives solutions. Intelligent horizontal offloading frameworks can overcome the majority of the identified restrictions of distributed computational offloading solutions. In summary, future research should incorporate the intelligence horizontal offloading system across a hybrid strategy to obtain the most effective and efficient intelligence offloading framework, independent of the specific protocol being targeted.

Table 5 A summary of the literature on intelligent horizontal offloading

Category	Reference	Architecture				Protocol						Strategy		Operating type			Objectives				Applied approaches	Evaluation tools
		Pattern										Resource allocation										
		P2P	Hierarchical	Publish/sub	Client/server	Gossip	Blockchain	Paxos	Zab	Raft		Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Intelligent	[89]	✓											✓		✓		✓	✓			FL	SI
	[90]	✓							✓				✓		✓		✓	✓			FL	SI
	[91]				✓								✓			✓			✓		DRL	PY
	[92]				✓					✓			✓		✓			✓	✓		DNN	SI
	[93]				✓								✓		✓		✓	✓	✓		DNN	PY
	[94]			✓									✓	✓			✓	✓			DRL	PY
	[95]		✓										✓	✓			✓			✓	CNN	SI
	[96]		✓							✓			✓		✓		✓	✓			DRL	SI
	[97]	✓								✓			✓				✓			✓	-	PY
	[98]				✓								✓		✓			✓	✓	✓	AID	MNW
	[99]	✓								✓			✓	✓			✓				DRL	SI

5.2 Vertical offloading

5.2.1 Rule-based

Liu et al. [100] proposed a pricing-based distributed offloading framework with constraints on computational capacity in the context of mobile-edge computing, where each user device is responsible for determining whether a newly received task should be processed locally or offloaded to a cloud node. The researchers employed a Stackelberg game to model the interaction dynamics between cloud servers (CSs) and edge servers (ESs) and the users' devices, designating the edge cloud as the leader and the users as followers [101, 102]. The methodology involves the edge-cloud setting prices for users' CPU cycles, after which the users decide on local processing or offloading based on the announced prices. They employed the Zab protocol to broadcast consistent pricing to every user in the edge cloud; however, the model uses a client/server architecture to communicate with users. The static channel model is taken into consideration, in which the channels do not altfog clouder during each offloading session. In this approach, each user autonomously makes offloading decisions to minimise latency and cost. Users then transmit their offloading decisions to the edge cloud for price adjustments. Similarly, Mitsis et al. [103] also introduced a model incorporating price and risk considerations into decision-making for data offloading in edge computing systems. Despite following the same Stackelberg game-based approach, this model distinguishes itself by addressing users' risk-aware decision-making in light of the potential overexploitation of MEC servers. They aimed to enhance the realism and comprehensiveness of the decision-making process in a multiuser-multiserver-multiaccess edge computing setting. In their approach, the offloading decision-making mainly depends on the value of the normalised effective demand communicated by servers to users, an indicator of MEC server exploitation levels. The communication pattern among distributed servers (leaders) and the users (followers) is client/server, the protocol, however, is Raft.

Xiao et al. [104] also modelled the problem using a Stackelberg game aimed at maximising the utility between the mobile edge platoon cloud (MEPC) and task vehicles. They proposed a mobile edge platoon cloud (MEPC) framework to dispense dynamic resources to proximate task-oriented vehicles promptly and responsively. Their formulation consists of task vehicles as the leader and the platoon members (PMs) as the followers. The task vehicle (leader) establishes a series of service pricing, prompting the PMs (followers) to adjust their resource allocation decisions to optimise their utilities by the specified service pricing. The communication protocol among high-performance vehicles as leaders and members is Raft. Similarly, Xu et al. [105] presented a distributed offloading technique based on a Mobile Edge Computing (MEC) network, in which the revenue of the edge cloud is modelled as a maximisation issue and the quantity of offloaded jobs for each user is a constraint-based on a static channel model, in which the channel stays constant throughout each offloading cycle but may change during other cycles. They used the Stackelberg game to track how users interacted with the edge cloud. The edge cloud uses the Zab protocol to choose which users to offload, assign bandwidth to them, and broadcast it to the offloaded users. The idea of distribution is derived from the bandwidth-sharing algorithm and the completely local computing method's optimised bandwidth allocation. Reduced latency in distributed offloading is the outcome of these reactive works that rely on distributed technology for decision-making.

Xu et al. [106] proposed a blockchain-based approach to computation offloading in edge computing within the context of 5G networks. This approach redistributes tasks to new edge nodes (ENs) for load balancing. This involves the utilisation of a consortium blockchain to facilitate computation offloading–operating within the Internet of Vehicles framework–with the selection of well-balanced strategies. Then, They follow an advanced evolutionary algorithm like the Nondominated Sorting Genetic Algorithm (NSGA-III) for solving multi-objective optimisation problems. This algorithm operates in a reactive manner, recording new offloading transactions as blocks that are appended to the blockchain for use in future decision-making. This work operates reactively based on distributed technology for decision-making, resulting in decreased latency in distributed offloading.

Slim et al. [107] and Mahapatra et al. [108] proposed a threshold-based offloading strategy, in which a distributed data centre system considers the location of requests and when a request arrives in the system, it is blocked by the first data centre along the data path. Slim et al. [107] employed a game-theoretic resource allocation approach involving two player classes, those with stringent latency requirements and those with delay-tolerant attributes, competing for resources across different data centres. A key distinction between the proposed algorithm and centralised ones lies in the extent of information exchange among the edge data centres and the central dispatcher. They also utilised the Tricircle approach of OpenStack, which is based on local information. The communication protocol and pattern are unclear. Mahapatra et al. [108] used a Fuzzy logic algorithm for determining the target layers for offloading considering the resource heterogeneity and the system requirements like network bandwidth, task size, resource utilization and latency sensitivity. IoT-driven collaborative real-time architecture with hierarchical information utilising

Fog and Edge, however, the protocol is not clear. Peng et al. [109] also introduced a distributed threshold-based offloading strategy in collaborative edge-cloud computing environments, employing three constrained multi-objective evolutionary algorithms (CMOEAs) to address multi-objective computation offloading problems. They presented the system hierarchical model composed by cloud servers, edge servers, and mobile devices (MDs), which can communicate with the edge server with a wireless link. Their methodology combines population-based search algorithms with adaptable constraint-handling mechanisms. A penalty function is common in their approach to appropriately manage objectives and constraints by a predefined penalty factor into each objective that lead to transforming a constrained multiobjective optimisation problem (CMOP) into an unconstrained MOP by integrating an overall constraint violation multiplied. Both studies exhibit reactive behaviour determined by the dynamically updated threshold value in the decision-making process.

Kaur et al. [110] and Zhao et al. [111] introduced two specialised task scheduling approaches and mathematical models for distributed offloading to minimise offloading delay within heterogeneous environments. Kaur et al. [110] proposed a hierarchical fog-cloud real-time scheduling algorithm, which schedules jobs on fog/cloud processors while ensuring their deadlines are met. They applied the Least Laxity First (LLF) heuristic algorithm to identify the most suitable fog node for scheduling. They aim to reduce delay by assigning jobs to preferred fog nodes to meet deadlines for regular tasks, which may be distributed across multiple fog nodes to ensure timely completion, or assigned to a preferred node if splitting is unnecessary. The decision-making process takes place at the fog level in a distributed manner, where each fog node has a network connection capable of executing jobs with minimal latency. The communication pattern among distributed nodes follows a hierarchical structure. To complement the theoretical insights, Zhao et al. [111] validated his approach through practical deployment. They proposed a task scheduling and resource allocation strategy in heterogeneous cloud environments to minimize delay within the realm of mobile edge computing. They employed the Gradient Descent algorithm, an optimisation technique used to determine the minimum value of a function. They observed distinct optimal offloading policies under scenarios of heterogeneous cloud and edge-cloud when the network experiences heavy traffic in both simulation (to demonstrate scale-up) as well as a lab-based testbed. In a multi-user case, the system must determine how computational resources in the edge cloud are allocated to users and how tasks are scheduled across heterogeneous cloud environments. Both these studies exhibit a reactive approach based on the diverse nature of decision-making within heterogeneous environments.

Wang et al. [112] proposed a distributed dynamic cluster-based offloading strategy within the context of vehicular edge computing, concentrating on a mixed computation offloading scenario involving mobile edge servers (i.e., vehicles), static edge servers, and the cloud server as resource providers. Their methodology entailed the utilisation of an offline Bound-and-Bound based Optimal (BBO) algorithm for periodic scheduling, taking into account the overall tasks' requirement statistics. Due to the limited information available to end users regarding the surrounding computation resources, roadside units (i.e., RSUs) run a low-complexity online heuristic algorithm to facilitate real-time—reactive–offloading decisions upon the arrival of each computation task. Each edge device updates its offloading decision, enabling the online heuristic algorithm to schedule tasks based on the current computation offloading environment in the ongoing scheduling period. According to the Zab protocol, every vehicle broadcasts a message to all of its neighbouring vehicles that includes the connection time and the computation resource, however, the communication pattern is unclear. Zhao et al. [113] also focused on the vehicle cloud computing environment. They proposed a priority-based task-offloading algorithm, which was implemented in the domain of vehicle cloud computing (VCC) framework. In their approach, performance vehicles shared computing resources to support vehicles unable to complete data processing within the specified time frame. They classified running vehicles into two categories: task vehicles and service vehicles. The task vehicle transferred the computing task to the service vehicle under the supervision of the edge server. They developed a Newton iteration algorithm on the MEC server that is independent of resource consumption, resulting in rapid offload finding. The communication pattern and protocol among vehicles and edge nodes are not clear. Both these works operate in a decision-making approach that is continuously updated iteratively.

Xin et al. [114] proposed a computing load balancing task offloading (CLB-TO) mechanism in the realm of Multi-access edge computing (MEC). This approach dynamically offloads each task that arrives at multi-access edge computing systems. They utilised two heuristic algorithms: the transmission delay-based CLB-TO and the genetic algorithm-based CLB-TO. Their methodology addresses scenarios where certain edge servers are overloaded with tasks while others remain underutilized leading to task blockages on the heavily loaded servers due to insufficient computing resources. The concept of a distributed mechanism is from the aspect of the interconnected metro optical network that employs the software-defined networking (SDN) controller to improve latency. The communication pattern between the

distributed edge servers is client/server the protocol, however, is unclear.

Ataie et al. [115] focused on a fog-cloud IoT system. They proposed a distributed threshold-based offloading strategy, where fog nodes act as forwarders, directing incoming requests to their optimal neighbours or the cloud or computing nodes that handle scheduling and forwarding. In their approach, they considered several metrics including QoE, security, task deadlines, and network resource utilisation. Their algorithm functions consist of task acceptance, rejection, or offloading based on deadlines and memory requirements. It utilised hop values–some nodes that a task can pass through before reaching its destination–to manage network resources and alleviate congestion while monitoring link bandwidth and dynamically tagging task workloads. Their approach allows some fog nodes to work as computing nodes to handle requests locally thereby reducing reliance on network resources. They also implemented the Gossip protocol for the selection of one node to handle notification and communication between computing and forward nodes the protocol, however, is peer-to-peer. Lastly, Adhikari et al. [116] also introduced a distributed threshold-based offloading strategy in a hierarchical fog-cloud placed in a static environment, where IoT applications utilised a multilevel-feedback queueing model. Similar to Ataie et al. [115] an IoT gateway receives each task to assign a priority based on its deadlines. The task priority then determined the allocation of the appropriate multilevel feedback queue, thereby diminishing the waiting time for delay-sensitive tasks in the queue and alleviating the starvation issue of low-priority tasks. In their approach, owing to the constrained processing and storage capabilities of IoT devices, the majority of tasks are dispatched based on the Raft protocol to locally distributed fog devices or centralized cloud servers, leveraging the task deadline and communication delay between the IoT device and computing device (fog or cloud server) for making offloading decisions.

Shakkeera et al. [117] introduce a hybrid system, incorporating deep reinforcement learning, game theory, and blockchain to optimise mobile data offloading in mobile cloud computing environments. The proposed approach employs a Deep Deterministic Policy Gradient (DDPG) agent to generate flexible offloading strategies to ensure fairness, efficiency, and security. Then a game-theoretic reward mechanism is regulated to encourage users based on predefined contribution rules. In their approach, blockchain technology is employed to record all offloading-related operations, providing permanent storage and access control in order to ensure data integrity and privacy. The proposed model is implemented over a mobile cloud environment with an evaluation across different network configurations, showing a special improvement in data offloading efficiency, reduced latency, and robust protection against malicious behaviours. Although DDPG introduces learning-based adaptability, the overall control and incentives remain grounded in rule-based governance in mobile cloud computing.

Gupta et al. [118] presents an intelligent resource optimisation for scalable and energy-efficient based on metaheuristic algorithms in the context of heterogeneous IoT devices, where a full-stack architecture designed to handle resource-constrained and diverse IoT devices through a hierarchical and adaptive structure. In their approach, an African vulture's optimisation algorithm (AVOA) is employed to establish connections between Cluster Head (CH) nodes. In order to minimise energy consumption and latency in scalability environments, strategically selecting edge servers based on proximity and performance is proposed. In terms of scalability, a dynamical system is proposed to adapt to changing environmental conditions and communication demands, leading to scalable deployment across complex IoT scenarios. The concept of scalability and latency in the real world comes from a simulation set-up, making a comprehensive Multi-Edge-IoT system within a vast coverage of real-world environments to improve energy efficiency, communication overhead, latency, and packet loss rate. The communication protocol is P2P.

To bridge the gap between theory and practice, Gedawy et al. [116] introduce concrete implementation in the field of computation offloading which is an open-source framework designed to support the development, testing, and evaluation of edge computing applications and scheduling algorithms. The Cumulus platform's suggested architecture consists of three core components, such as a testbed with a central task dispatcher, a diverse set of standalone IoT devices, and a controller that manages a local IoT cloud to simplify task distribution. In their approach, a web-based live visualiser is employed to enable real-time monitoring of devices for experimental and research purposes. Therefore, their findings highlight the practical challenges and insights of real-world deployment, emphasising the need for efficient device-to-device communication protocols and demonstrating how system heterogeneity can significantly impact performance-factors that are often overlooked in purely theoretical studies.

This subsection discussed rule-based vertical offloading approaches based on their specific characteristics. The most common architectural patterns in this category are hierarchical and client/server patterns, while P2P is rarely used (e.g. [115]). In Table 6, rule-based approaches are summarized based on architecture, strategy, and key objectives. Although Gossip and Blockchain protocols appear in a limited number of studies, other protocols like Zab and Raft are mainly used. Performance and reliability remain the main optimisation objectives, but fewer approaches address

Table 6 A summary of the literature on rule-based vertical offloading

Category	Reference	Architecture									Strategy					Objectives				Applied approaches	Evaluation tools
		Pattern				Protocol					Resource allocation		Operating type								
		P2P	Hierarchical	Publish/sub	Client/server	Gossip	Blockchain	Paxos	Zab	Raft	Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Rule-based	[100]				✓				✓		✓	✓	✓			✓		✓		GT	SI
	[103]				✓					✓		✓	✓					✓	✓	GT	SI
	[104]			✓						✓			✓			✓				SE	SI
	[105]			✓					✓		✓		✓			✓				SE	SI
	[106]		✓				✓						✓			✓				BC	SI
	[107]											✓	✓			✓			✓	GT	OS
	[108]		✓									✓	✓			✓				FLA	IFS
	[109]		✓									✓	✓							OA	SI
	[110]		✓										✓			✓				OA	RW SI
	[111]		✓							✓			✓			✓				OA	SI
	[112]								✓			✓	✓			✓			✓	HA	PY
	[113]										✓		✓			✓				OA	SI
	[114]				✓							✓	✓			✓			✓	HA	SI
	[115]	✓				✓						✓	✓			✓			✓	OA	OMN
	[117]						✓					✓			✓	✓			✓	GT DRL	SI
	[118]	✓									✓		✓			✓	✓		✓	HA	SI
	[116]		✓							✓	✓					✓				HA	SI

cost and reliability. Several tools and applied approaches such as SE, OA, and HA, alongside evaluation tools like SI, RW, and IFS, have been used (e.g., [105] and [110]). In summary, verticalZab rule-based approaches provide high performance and reliability offloading, but future enhancements should focus on developing energy-aware hybrid strategies to improve adaptability and efficiency.

5.2.2 Optimised

Liu et al. [119] approach combined the alternating direction method of multipliers (ADMMs) with particle swarm optimisation (PSO) to address the dynamic task offloading (PTO) problem. The PTO aims to reduce the weighted total of payment costs, energy use, and execution delays. They considered the diverse characteristics of vehicular fog (VF), fog server (FS), and central cloud (CC) computation servers. In their methodology, the FS was tasked with deciding on task offloading during a scheduling interval for vehicular requesters within its service area, where tasks can be fragmented and allocated to different server types. The fog server (FS) will use Zab protocol to broadcast the control message in its service region. The distributed nature of their approach stemmed from each FS—representing the VF set in a specific area—functioning as a local scheduler overseeing scheduling decisions for pending tasks in its respective service domain. The interaction between end-users and computation servers during task offloading could be facilitated through various wireless channels, such as V2I and V2V communication. Gao et al. [120] also combined Particle Swarm optimisation (PSO) and a hybrid heuristic algorithm for joint static service caching and task offloading in the context of edge-cloud computing, wherein numerous user devices established communication links with an edge. They employed a joint service caching and task offloading technique, in which the service caching solution was responsible for determining which tasks were suitable for offloading to edges. In their methodology, the decision-making process for task offloading is mainly based on how well each solution by a particle performs to improve the overall performance. The particles are updated using a lévy flight movement for the best overall solution, the global best, and the best solution for each particle, the local best. The communication pattern and protocol among cloud and edge nodes are unclear.

Roy et al. [121] also introduced a distributed resource management system in the context of a dew-based edge-to-cloud computing environment that is a programming model for enabling plug-in facility-empowered personal networks in distributed computing through P2P communication [122]. In their approach, a Control Access Unit (CAU) was tasked with allocating resources to service requests in one of the three computation layers, considering various Quality of Service (QoS) factors. They followed a hybrid adaptive metaheuristic algorithm, which combined learning-based adaptive particle swarm optimisation and genetic algorithms. In their approach, the challenge of dynamic and unpredictable offloading decision-making, which frequently changes over time, was addressed by employing multiple velocity updating strategies to mitigate premature convergence issues. The decision-making occurs at the edge level, with the CAU playing a key role in implementing a distributed management strategy that optimizes resource selection based on service requirements. The communication pattern between the distributed edge servers and IoT devices is client/server the protocol, however, is Raft.

Li et al. [123] presented a framework for a distributed multi-leader single-follower Stackelberg game focused on task offloading and resource allocation within the realm of Internet of Vehicles, where each edge server (ES) engaged in both cooperative and competitive interactions with other servers, collaborating on tasks while also competing with each other to increased profits. They followed a Karush-Kuhn-Tucker (KKT) algorithm to address a multi-objective optimisation challenge aimed at simultaneously reducing task processing delays and costs. Their methodology involved making offloading decisions based on current state information and transmitting these decisions to the task vehicles (TV) through broadcasting based on Raft protocol that subsequently divides tasks and executes locally assigned subtasks. Throughout the offloading process, the communication between the TV and service vehicles (SVs) is achieved by frequency division multiplexing access (FDMA). Cheng et al. [124] introduced a multi-user (client) and multi-server mobile edge computing (MEC) network to address dynamic computation offloading and resource allocation, however, with the same objective, i.e., within the context of integrating MEC systems with Unmanned Aerial Vehicles (UAVs). The communication pattern is client/server. They employed a Markov Decision Process framework, treating the joint sequential problem as a Partially Observable Markov Decision Process (POMDP) and presenting an online Multi-Agent Proximal Policy Optimisation (O-MAPPO) scheme to manage the sequential decision-making challenge to maximize the energy efficiency. In their strategy, all ground equipment (GE) and mobile edge computing (MECs) work as distributed agents, with each agent making decisions based on partial observations and sharing a common reward. Both these works operate in a reactive nature based on the multi-user environment. The communication protocol for both works is Raft.

Wu et al. [125] and El Haber et al. [126] have developed energy-aware Edge-Cloud collaborative computation offloading solutions. Wu et al. [125]'s algorithm

considered the computing and communication resources of mobile devices, edge servers, and cloud servers for offloading, dynamically producing an offloading strategy. On the other hand, El Haber et al. [126] consider network operators (NOs) leasing resources from a high-tier central cloudlet for task offloading. The authors jointly minimize the NO's computational cost and devices' energy consumption in a multi-tier MEC system by optimising offloading decisions, transmission power allocation, radio resources on the uplink channel, and assigned server computation, all while respecting device latency requirements. Wu et al. [125] employed a nonlinear exponential inertia weight particle swarm optimisation (PSO) algorithm for solution generation, whereas El Haber et al. [126] mathematically formulated a mixed-integer non-convex program and introduced a Branch-and-Bound (BnB) algorithm to find the optimal solution. While both Wu et al. [125] and El Haber et al. [126] improve energy consumption and cost, El Haber also aimed to enhance latency. Both these works operate in a reactive nature. The communication pattern and protocol in both works are unclear.

Xiao et al. [127] adopted a collaborative approach of task offloading for mobile edge computing (MEC) within small cell networks, where each small cell (SC) was equipped with an edge server (ES) to serve the user devices (UDs) within its coverage area. They employed a structural technique to mitigate the strong interdependence among various variables in the problem. Their methodology revolves around addressing three distinct subproblems, including the offloading decision subproblem, the radio access network (RAN) spectrum allocation subproblem, and the joint distributed wireless backhaul (DWB) spectrum and computation capacity allocation subproblem. The concept of a distributed mechanism comes from the distributed wireless backhaul (DWB) in which some small cells (SCs) serve as gateways, connecting to the core network through fibre-optic cables. Tasks sent to these SCs first go through the gateway using DWB, before being transmitted to the remote cloud data centre (CDC) via fibre-optic links. Bi et al. [128] proposed a distributed energy-aware task offloading strategy, however, with the same environment and objective, i.e., they utilized an optimisation algorithm within the context of heterogeneous networks, where the concurrent execution of tasks on mobile devices (MDs), macro base stations (MBSs), and/or small base stations (SBSs). They followed particle swarm optimisation combined with genetic learning (PGL), merging the robust local search capabilities of a particle swarm optimizer with the genetic operations of a genetic algorithm. In their approach, the decision-making process for offloading was primarily based on formulating a constrained mixed integer non-linear program (MINLP) aiming to optimize offloading choices and reduce overall energy consumption within a mobile edge computing (MEC) system. The distributed nature of this mechanism is characterized by computation taking place simultaneously at MDs, MBSs, and SBSs. While the communication pattern among mobile devices is client-server, the specific protocol is unclear. While both Xiao et al. [127] and Bi et al. [128] improve energy consumption, Xiao also aimed to enhance latency. In terms of security, Bi et al. [128] propose a blockchain-based and trust-aware task offloading strategy (BBTAS), where tasks are treated as blockchain transactions. Additionally, these transactions are automatically and securely executed by smart contracts (SCs). The proposed approach is adopted a utility-based decision-making method (UB-DMM) to drive offloading decisions, which supports structured procedures for resource allocation, trust computation, and verification of both service requestors (SRs) and service providers (SPs). In their approach, an RFM-TPM-based trust method is introduced to effectively protect against network insider attacks, such as varied recommendation attacks and service attacks, and the trust proof may be securely stored on the blockchain. In terms of experiment, smart contracts (SCs) automatically execute predetermined actions, producing verifiable data that is distributed throughout a peer-to-peer (P2P) network and validated on the blockchain. As a result, the suggested approach intends to boost time efficiency by adding trust qualities into the blockchain, while also improving resilience against various security assaults through a full security analysis.

Rahmani et al. [129] propose the hybrid A3C-AHP framework, which integrates Asynchronous Advantage Actor-Critic (A3C) reinforcement learning with the Analytic Hierarchy Process (AHP) for multi-factor decision-making in the context of blockchain-enabled Mobile Edge Computing networks. The first part of the proposed approach (A3C) enables IoT agents to learn adaptive offloading strategies by interacting with dynamic network conditions, while the second part (AHP) provides a structured method to prioritize criteria such as energy consumption, latency, and trust. In their proposed approach, a peer-to-peer (P2P) blockchain network can help the framework to ensure secure task validation and transparent data transmission among computing nodes. This approach enables a dynamic and flexible offloading strategy. Unlike previous theoretical models, the scalability of the A3C-AHP solution has been carefully evaluated through simulations involving up to 100 computing nodes and a thousand IoT, demonstrating its real-world applicability. Therefore, the proposed hybrid approach improves energy efficiency, latency, and computational overhead, directly addressing concerns related to battery-constrained IoT devices in the real world. Moreover, combining the distributed development and asynchronous

structure scalability of the algorithm helps the system to maintain stable performance in high-density areas.

Wang et al. [130] introduce a joint optimisation framework called User Association and Computation Offloading (UACO) in order to minimise total energy consumption while satisfying offloading rate constraints. The proposed method considers the problem a mixed-integer linear programming (MILP) problem that makes it NP-hard, using an Offload Exact Constrained Potential Game (OFECPG) based on game theory to facilitate distributed execution. In their approach, the best response dynamics are based on local decomposition (BR-LD), and better response dynamics are based on the local switch operator (BR-LS) to enhance computational efficiency. Therefore, the utilisation of game-theoretic dynamics and constraint-aware optimisation makes this method a strong candidate in the vertical optimised category, particularly for large-scale and mobile edge scenarios. The communication pattern between the distributed edge nodes is client/server the protocol, however, is unclear.

Liu et al. [131] proposes a statistical method for dynamic task offloading and resource allocation in the context of mobile edge computing under ultra-reliable and low-latency communication (URLLC) restrictions. Their approach relies on task queue lengths by utilising extreme value theory to limit users' power consumption while trading off the allocated resources for local computing and task offloading. Moreover, the proposed two-timescale optimisation system dynamically impacts user-server associations and task offloading in response to real-time channel situations, server workloads, and queue states. This study indicates that considering higher-order statistics and unpredictability in network dynamics resulted in substantially more reliable and delay-efficient task processing in actual, heterogeneous MEC situations.

Alameddine et al. [132] also mathematically define and formulate the dynamic task offloading and scheduling problem (DTOS) as a mixed integer program (MIP) (DTOS-MIP) in the context of multi-access edge computing (MEC). In order to efficiently solve the DTOS problem, a logic-based benders decomposition (LBBD) is proposed that decomposes the original problem into a master problem, responsible for task offloading and resource allocation, and multiple sub-problems, each handling the scheduling of tasks assigned to individual IoT applications. The proposed method significantly decreases computational latency and explains how optimised scheduling and offloading decisions improve latency performance across a wide range of QoE requirements associated with 5G vertical industries. This framework focuses on both resource allocation and fine-grained offloading in IoT applications, which have direct effects on QoE improvements. In their user-centred design, both the practicality and acceptance of offloading systems in real-world deployments considerably are improved by positively impacting offloading decisions and increasing the number of tasks admitted to the network. Zhou et al. [133] also present a comprehensive framework that prioritises reliability-aware and user-centric Quality of Experience (QoE) in multi-user Mobile Edge Computing (MEC) systems. Their approach explicitly considers both transmission and execution reliability in the optimisation process. To manage the complexity of task offloading and scheduling, the problem is decomposed into three interrelated sub-problems: task-to-edge server assignment, resource allocation, and offloading decision-making. For each of these, the authors design a reliability-aware task-to-ES assignment algorithm, a QoE-optimal resource allocation strategy, and a binary particle swarm optimisation (BPSO)-based task offloading model. QoE is quantitatively modelled using a system utility function, ensuring user satisfaction is directly embedded in the optimisation process. Moreover, the proposed framework incorporates a reliability-optimality theorem for task-to-ES assignment based on the model fault rate and the exponential distribution. As a result, both simulation and real-world testbed experiments indicate the effectiveness of the proposed method, demonstrating significant improvements in average QoE compared to benchmark approaches.

Yu et al. [134] Introduce a new dynamic mobility-aware partial offloading (DMPO) decision scheme in the context of mobile edge computing. In their approach, after movement, the user equipment (UE) may change not only the serving base station and virtual machine (VM) placement but also the proportion of a task to be offloaded. In terms of mobility prediction, they proposed a unique integrated algorithm for partial offloading and communication path selection, aiming to minimise energy consumption while satisfying the execution delay constraint. This algorithm predicts the movement between two base station handovers and determines the offloading proportion for each time slot according to the network conditions, remaining task load, and latency constraint. The simulation is implemented in Java, comprising a single UE, a base station, and an SDN controller, which communicate through sockets. The results demonstrate improved performance in satisfying the delay constraint compared to conventional approaches.

There are fewer works presented in this subsection compared to Sect. 5.2.1, indicating that research on optimised vertical offloading frameworks is still in its infancy. Many of the offloading solutions described in this subsection claim a form of raft protocol support. However, some domains are either not modelled or supported with any protocol (e.g., [127] and [128]). Table 7 summarizes the shortcomings and open issues discussed in Sect. 5.2.2. Many of the challenges associated with hierarchical and public/sub patterns still

Table 7 A summary of the literature on optimised vertical offloading

Category	Reference	Architecture									Strategy					Objectives				Applied approaches	Evaluation tools
		Pattern				Protocol					Resource allocation		Operating type								
		P2P	Hierarchical	Publish/sub	Client/server	Gossip	Blockchain	Paxos	Zab	Raft	Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Optimised	[119]	✓							✓			✓	✓			✓	✓	✓		HA	SUM
	[120]										✓			✓		✓				HA	SI
	[121]	✓			✓					✓		✓			✓					HA	CSP
	[123]									✓			✓			✓		✓		GT	SI
	[124]									✓		✓	✓				✓			MM	SI
	[125]											✓	✓				✓			HA	SI
	[126]												✓			✓	✓	✓		OA	SI
	[127]											✓								OA	MT
	[128]				✓												✓			HA	RW
	[128]	✓					✓						✓		✓	✓			✓	BC	PY
	[129]	✓					✓					✓			✓	✓	✓		✓	RL AHP	SI RW
	[130]				✓							✓	✓				✓			NE	SI
	[132]											✓	✓			✓				MIP	RW
	[133]				✓						✓		✓			✓			✓	HA	SI RW
	[134]				✓							✓			✓	✓	✓			HA	SI

apply to optimized vertical offloading solutions. In summary, as the enabling technologies (e.g., MM [124] and GT [123]) mature and become more sufficient, it will be easier to implement more efficient frameworks using those approaches for future enhancements.

5.2.3 Intelligent

Robles et al. [8] approach followed a Reinforcement Learning (RL) algorithm to address the Task Assignment Problem (TAP) at the edge layer to dynamically reduce energy consumption and execution time. In this approach, each edge device is an RL agent that decides the destination for task execution, i.e. whether local, fog/edge or cloud layer. They employed a novel multi-layer extension of RL (ML-RL) techniques that empowered edge agents to consult an agent at the upper layer to obtain knowledge for performance improvement in intricate and unpredictable scenarios. In their methodology, the RL agents of the devices could assign the offloading decision to an agent of a higher layer based on a heuristic value calculated for each layer. Additionally, they broadened the scope of the RL algorithms to a fresh multi-layer strategy that enables the agent to transfer the offloading decision to another agent. The learning procedure involves the utilisation of a Q-value table, which is utilized to retain and retrieve the value of the Q-function for each state and action to update the new Q-value in the table. The communication pattern between the distributed edge nodes is client/server the protocol, however, is unclear

Xun et al. [135] proposed a distributed delay and energy-conscious task scheduling system for edge clouds, which aims to achieve optimal scheduling at each edge node without the need for a detailed mathematical model of the environment. They employed a Deep Q Network-Action (DQN-A) algorithm, which integrates a deep Q-learning network with a priority-based action selection approach. Their methodology addresses the challenge of estimating the load status of edge nodes in dynamic settings and identifying nodes that can ensure the necessary quality of service—formulated using a Markov decision process (MDP)—while avoiding congestion. The communication pattern and protocol among devices and edge nodes are not clear. However, Dai et al. [136] created a unique Lyapunov-guided deep reinforcement learning (DRL) for delay-aware dynamic offloading method (LyD2OA) for mobile edge computing (MEC) devices, where they established predefined queue length thresholds. They used a Lyapunov optimisation theory that relies on additional uncertain knowledge and reformulates it into a new issue that is only dependent on the state of the system at the moment. In order to achieve asymptotically optimal solutions, they used Lyapunov optimisation, a stochastic optimisation technique that permits the decoupling of spatiotemporally coupled variables. The distributed technique of this system is defined by the task offloading choice for the slot that still depends on local slot information. Although the pattern of communication between mobile devices is published and subscribed, the specific protocol is unclear. Both these works operate in a proactive nature to improve energy consumption and delay.

Zhu et al. [137] approach is based on a multi-agent deep reinforcement learning (DRL) technique for task offloading for mobile-edge computing (MEC) focusing on the vehicular computation offloading issue. Their proposed strategy included a centralized training algorithm and distributed execution algorithm based on multiagent DRL, where the data center is responsible for training the actor-network and critic network in a centralized fashion and each vehicle is a distributed agent making autonomous offloading decisions. The communication pattern and protocol among vehicles and BS are unclear. Similarly, Liang et al. [138] also focused on the vehicle scenario. However, in this case, each distributed agent (vehicle) operates on local data to enhance overall system performance. They employed a fingerprint-based deep Q-network approach that lends itself well to distributed deployment. This approach focused on ensuring reliability in the periodic sharing of safety messages over V2V links, adapting V2V spectrum sub-band selection and power control in response to localized channel variations occurring within the message transmission timeframe. The base station (BS) estimated the channel for each time slot, after which it broadcasted to every vehicle within its coverage using the Zab protocol. Each device receives information about edge server utilisation via a V2V communication mechanism, which can be facilitated through cellular (Uu) and sidelink (PC5) radio interfaces. Both of these works followed a proactive approach to decision-making and adapted to the dynamic and unstable environment. In both works, they aim to reduce the task execution delay over an extended period of time.

Rahbari et al. [139] focused on mobile fog computing (MFC). They proposed a placement approach that utilises the classification and regression tree (CART) to identify the optimal fog device (FD). In their approach, the decision-making process for offloading primarily relied on the values of the CART. These values are continuously adjusted based on the average speed and cost parameters, which take into account key system components like processor (P), memory (M), storage (S), and bandwidth (BW) aspects as part of the module placement strategy. The proposed method refers to a distributed system where decision-making happens at each fog device. The goal is for each fog device to choose the best-suited fog device (FD) for specific tasks. To do this, each fog device updates its threshold values according to the range of values for decision parameters, which helps in selecting the

right FD. Gao et al. [140] utilised multiagent reinforcement learning for vehicular fog computing, where each mobile vehicle (MV) acted as an autonomous agent responsible for offloading decision-making. They introduced a distributed method for task offloading rooted in transformer and policy decoupling-oriented multiagent actor–critic (TPDMAAC). In this approach, each MV agent - likely a type of virtual machine or processing unit–independently chose the optimal computing node and assigned the necessary resources for incoming tasks. The concept of a distributed mechanism is from the distributed execution (CTDE) algorithm wherein each MV agent implemented distributed decisions regarding dynamic JTORA policies using the trained TPDMAAC model. Both of these works followed a proactive approach for task offloading. While both Rahbari et al. [139] and Gao et al. [140] improve performance, Rahbari also aimed to improve power consumption and Gao reduced system costs. The communication pattern and protocol in both works are not clear.

Shinde et al. [141] approach is also for vehicular edge computing systems, where they aimed to optimise latency and energy consumption through a combined consideration of FL and computation offloading mechanisms in a dynamic environment. The methodology involved utilising High Altitude Platforms (HAPs)–a particular type of unmanned aircraft–as FL servers and supporting Vehicular Users (VUs) as FL clients to make informed offloading decisions and broadcast global parameter sets based on Zab protocol. During each offloading request, the VUs engaged in multiple FL iterations with the HAP to accurately allocate the offloading segment towards the designated Road Side Unit (RSU). The centralized FL servers aggregate updates from the entities to generate new global parameters, which are then transmitted back to the users. The interaction pattern between distributed VUs and RSUs followed a client-server model. Liu et al. [142] also utilised a similar FL-based methodology, however, as a collaborative intrusion detection mechanism. Their approach also integrates blockchain to facilitate efficient model training and secure data sharing. Both studies followed a proactive approach to decision-making within the dynamic vehicular domain. While both Shinde et al. [141] and Liu et al. [142] improve performance, Shinde also aimed to improve power consumption and Liu reduced computation cost.

Zaman et al. [143] proposed a novel task offloading framework incorporating lightweight user mobility prediction using machine learning approaches in the context of mobile edge computing. In their approach, the non-linear nature of the user mobility data is combined with restricted input features and short window size, which mandates a lightweight model. To address the limited computing power of edge devices, they use a single-layer artificial neural network (ANN) with a NAREX encoder equipped with a Sigmoid activation function, resulting in low computational complexity during training and testing models, thereby making it suitable for resource-constrained edge devices. In terms of mobility prediction, they design a neural network that trains on mobile user location data and then implements the trained model at the edge network controller (ENC). A multi-objective genetic algorithm (MOGA) is used to optimise edge server selection and reduce latency and energy consumption while optimising resource utilisation across the MEC system. In the simulation, they use the MobFogSim simulator to compare the proposed LiMPO framework to DMPO and TAOM, illustrating that their model exceeds the benchmarks in terms of latency reduction, energy efficiency, and resource utilisation.

Lastly, Fan et al. [144] proposed a hybrid offloading mechanism that aims to optimise jointly the dynamical adjustment of the decisions at two layers, i.e. edge-to-edge (E2E) and edge-to-cloud (E2C). Their focus is on the service function chain (SFC) architecture [145]. As a technical approach, they employed Deep Reinforcement Learning (DRL) techniques to formulate the optimisation problem for the maximisation of the efficiency of edge resources. The approach involved converting the system model and optimisation problem into Markov Decision Processes (MDP) and presenting a Deterministic policy gradient (TD3) based SFC E2E and E2C joint offloading algorithm to address the modelled MDP. The utilisation efficiency is increased and the convergence performance is superb in this work. The communication pattern between the distributed edge servers is E2E the protocol, however, is unclear.

This subsection introduces a limited number of works compared to Sects. 5.2.1 and 5.2.2 that intelligent vertical offloading is still a developing research area compared to rule-based and optimised approaches. The selected works are mixed by all structure patterns except hierarchical architectures. As shown in Table 8, dynamic resource allocation and proactive techniques are commonly used to enhance adaptability and decision-making efficiency. Most solutions prioritise performance and energy objectives, with increasing attention given to cost and reliability. The use of advanced learning-based approaches like DRL, FL, and RL (e.g. [142] and [144]) illustrates a trend in vertical offloading that incorporates intelligent decision-making. However, some solutions (e.g., [139] and [140]) still lack integration with established architectural models or protocols. In summary, future work should attempt to bridge these gaps by adopting more structured architectures and leveraging intelligent models such as DRL and RTA to develop reliable, energy-efficient, and context-aware offloading frameworks.

Table 8 A summary of the literature on intelligent vertical offloading

Category	Reference	Architecture				Protocol				Strategy		Operating type			Objectives			Applied approaches	Evaluation tools
		Pattern								Resource allocation									
		P2P	Hierarchical	Publish/sub	Client/server	Gossip	Blockchain	Paxos	Zab	Raft	Static	Dynamic	Reactive	Proactive	Hybrid	Performance	Energy	Cost	Reliability		
Intelligent	[8]				✓							✓		✓		✓	✓			RL	PES
	[135]											✓		✓		✓	✓			DRL	PY
	[136]			✓								✓		✓		✓	✓			DRL	SI
	[137]											✓		✓		✓				DRL	SI
	[138]	✓							✓			✓		✓		✓				RL	SI
	[139]													✓		✓	✓			RTA	CSP
	[140]											✓		✓		✓		✓		RL	SI
	[141]				✓				✓			✓		✓		✓	✓			FL	MT
	[142]						✓					✓		✓		✓		✓		FL	PY
	[143]				✓							✓			✓	✓	✓			HA ML	FS
	[144]	✓										✓		✓		✓			✓	DRL	TF

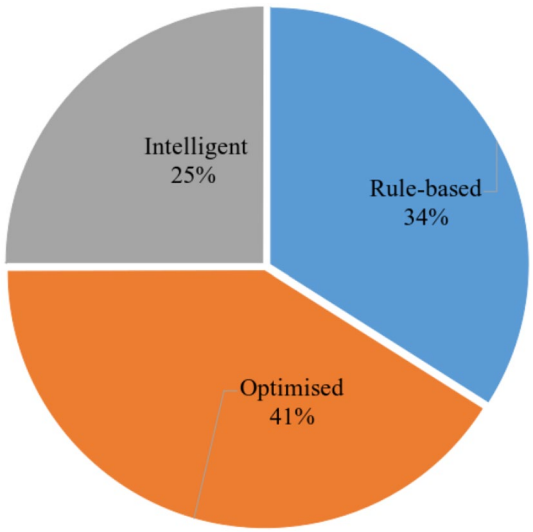


Fig. 6 Decision-making approaches in decentralised computation offloading solutions

5.3 Summary and analysis of reviewed solutions

After a detailed review of existing decentralised computation offloading solutions based on the proposed taxonomy, this section summarises the key findings. We present two illustrative charts that highlight the distribution of architectural characteristics and decision-making approaches across the surveyed studies.

Figure 6 summarizes the percentage of rule-based, optimised, and intelligent offloading approaches observed in the review section, based on the proposed taxonomy for offloading decentralized computation. As illustrated, optimised approaches constitute the largest percentage (41%), followed by rule-based methods (34%), while intelligent solutions represent a smaller share (25%). This distribution indicates that, although a significant focus has been placed on optimisation techniques, classical rule-based decision strategies are still extensively used, possibly due to their simplicity and lower computational requirements. Meanwhile, intelligent techniques, despite their promising potential for dynamic and context-aware decision-making, appear to be underexplored. Therefore, this can be an important opportunity for future research to further integrate AI-driven models into decentralised offloading frameworks.

As shown in Fig. 7, a significant proportion of decentralised computation offloading solutions adopt a P2P communication pattern (37.5%), indicating a strong preference for peer-to-peer architectures over hierarchical, client/server, and publish/subscribe models. Regarding protocol usage, consensus protocols such as Raft and Zab (each 14.8%) are more widely employed, whereas protocols like Paxos remain comparatively underutilised. In addition, dynamic resource allocation (68.2%) emerges as the predominant

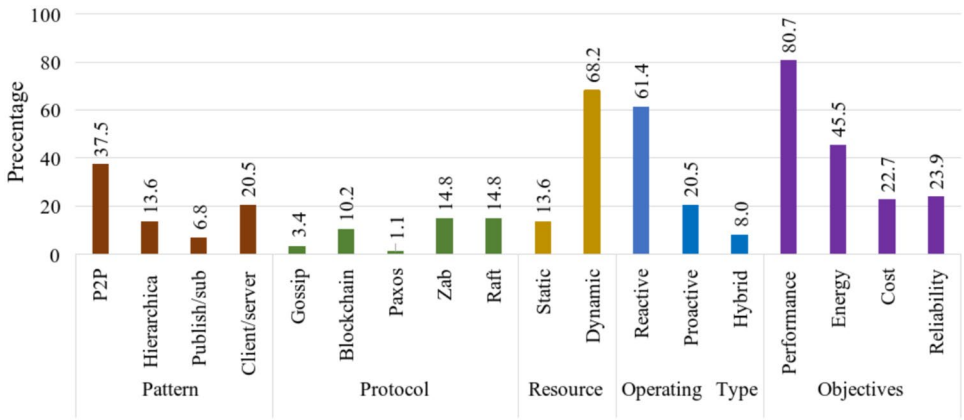


Fig. 7 Distribution of key characteristics in decentralised computation offloading solutions

strategy, highlighting the necessity for flexibility and adaptability in decentralised and dynamic computing systems. Moreover, reactive operating types (61.4%) dominate over proactive and hybrid schemes, reflecting a reliance on real-time system responses rather than predictive management. Concerning optimisation objectives, the majority of offloading frameworks prioritise performance (80.7%) and energy efficiency (45.5%) over cost (22.7%) and reliability (23.9%). These trends suggest that, although significant efforts have been devoted to performance optimisation and dynamic management, aspects such as fault tolerance, cost efficiency, and proactive resource management remain comparatively underexplored and offer promising directions for future research.

6 Issues and future directions

Section 2 provides a comprehensive review of existing survey papers in the field of computation offloading, as summarised in Table 1. Furthermore, Section 5 presents an in-depth review of individual offloading solutions, systematically classifying them into different groups based on a proposed taxonomy, with their key characteristics and research initiatives summarised by Tables 3, 4, 5, 6, 7, and 8. This section discusses and analyses the results presented in these tables to identify future directions for computation offloading in the edge-to-cloud environment.

6.1 Mobility in edge/fog

Despite some existing studies that consider mobility aspects like point-to-point and multi-user communications, mobility remains an open issue that requires further attention [146]. UAVs (Unmanned Aerial Vehicles), high-speed rails, moving hotspots, and 3D connectivity are some examples of upcoming applications that increasingly rely on ML algorithms to learn user behaviour and network dynamics. We believe that the growing complexity of mobility patterns in future networks will demand not only predictive but also adaptive offloading mechanisms capable of responding in real-time applications. Due to the inherent dynamic behaviour of mobile devices, which can rapidly move between various locations, there is a need for these devices to adapt and switch between dedicated servers distributed across a wide geographical area. In the context of high-mobility federated edge-to-cloud computing architectures, mobility awareness remains a significant challenge, partially regarding the transfer of tasks between different layers in distributed systems [147–149]. The majority of existing methods for task offloading in edge-to-cloud networks assume that users remain fixed during task offloading and that continuous communication between mobile devices and fog nodes is always feasible, an assumption that is impossible in real-world scenarios [150]. Fan et al. [151], for example, explore task scheduling with resource allocation for multiple users within a single-edge server, assuming universal access to the server anytime, which is an unrealistic assumption in real-world deployment. In our opinion, future research must not only model mobility more realistically but should also explore distributed coordination strategies among edge nodes to dynamically manage user movement without affecting service availability. Therefore, we highlight the need to address the challenges of mobility in mobile edge-to-cloud computing networks.

6.2 Intelligent resource allocation in edge/fog

A critical challenge in computation offloading is the accurate selection of the appropriate resources. Failure to do so may result in tasks being offloaded to distant locations in the event of a resource shortage following offloading [152]. This can lead to increased communication latency, and an elevated number of offloading hops may also compromise data privacy [153]. The utilisation of ML techniques can be beneficial in predicting the future availability of resources. Moreover, ML models can aid in forecasting the duration of each device's busy state during its processes, thereby

facilitating informed decisions for future offloading scenarios [154]. Although many research efforts have investigated resource management in fog and edge computing applications significant gaps remain. Some studies [155–157] discussed classic issues in resource management, including application location, task scheduling, resource provisioning, and load balancing. However, incorporating techniques, such as serverless computing, blockchain, quantum computing, and Software-Defined Networking (SDN) into intelligent resource management frameworks is mostly unknown. Moreover, most of the recent studies do not consider the Quality of Service-based approach to resource allocation methodologies. Future research should investigate how AI and ML techniques can be utilised to improve resource management in dynamic and heterogeneous fog/edge environments. They also need to consider current requirements for scalability, security, latency minimisation, and adaptability to new computing frameworks.

6.3 Energy prediction in edge/fog

Efficient task offloading remains essential for extending the operational lifespan of constrained edge devices; however, it is frequently associated with significant energy consumption challenges. A serious challenge is accurately predicting the amount of energy required during communication activities in task offloading. Although, machine learning (ML) models have been employed to predict energy requirements and consumption in some previous approaches [81, 158], most existing techniques lack real-time adaptability and lightweight design, making them unsuitable for highly dynamic and resource-constrained edge environments. Future efforts should focus on developing ultra-lightweight ML-based energy estimation models with minimal computational overhead and context awareness dynamically. Moreover, combining predictive energy models with proactive task scheduling and resource allocation might significantly improve the energy efficiency and reliability of edge-to-cloud offloading frameworks. From a broader offloading perspective, these aspects are essential and there should be more focus on incorporating both computational and transferring energy requirements.

6.4 Security

The following directions are based on the authors' critical reflections and insights gained during the preparation of this survey. In distributed offloading systems, ensuring user privacy and protecting both data and application confidentiality from adversaries play a crucial role in building and maintaining trust in mobile platforms. Security challenges in such environments can be split into two critical areas, such as mobile user security and data security [66, 128]. Mobile devices continuously connect with remote devices for distributed decision-making, leading to increasing vulnerability to various attacks, such as data interception, identity theft, and unauthorised access. There is still a crucial requirement for lightweight, adaptive security protocols developed for distributed offloading systems. Moreover, data security presents new challenges, particularly in protecting sensitive information transmitted between mobile and edge devices. The distributed offloading in the federated edge-to-cloud continuum contributes to these challenges since data is transferred in multiple layers and geographic locations, resulting in increasing vulnerability. Future research must prioritise designing resilient, distributed security frameworks that can dynamically adapt to the fluctuating and heterogeneous nature of edge-to-cloud offloading environments.

6.5 Lack of datasets/simulation environments

The availability of relevant datasets in the field of computational offloading is critical for conducting the comprehensive analysis and evaluation of different scenarios in the context of distributed environments. Based on my critical reflections and insights gained during the preparation of this survey, distributed edge-to-cloud offloading systems suffer from a lack of suitable datasets, incorporating a wide range of relevant parameters to offloading decision-making processes in both tasks and processing devices sides. For instance, response time, energy consumption, network latency, tasks' deadlines, resource utilisation, resource requirements, computational overhead, and user satisfaction are some of the necessary parameters for offloading decisions. Moreover, datasets must present various scenarios for different network conditions, task characteristics, mobility patterns, and offloading targets to facilitate a holistic understanding of offloading system performance. However, current publicly available datasets and simulation environments remain limited in both scale and realism, restricting the ability to benchmark and validate offloading strategies effectively for distributed scenarios. Future research should prioritise the development of open-access, standardised datasets and digital twin platforms that simulate heterogeneous edge-to-cloud on distributed scales involving multiple types of devices, distributed infrastructures, and communication technologies.

6.6 Interoperability/portability in edge-to-cloud

Interoperability refers to the vendor-specific tendency of systems and services in edge-to-cloud computing to work best within their ecosystem, often limiting seamless

integration with other architectures. This ability among heterogeneous platforms and service providers represents a critical challenge in IoT/edge/fog systems. Interoperability in IoT is challenged by the large range of communication protocols-such as Wi-Fi, Bluetooth, Zigbee, LoRa, 6LoWPAN, and CoAP-each protocol supports an individual IoT use case, no single protocol can meet every aspect of IoT requirements on its own [159]. Moreover, portability also represents the ability to seamlessly migrate applications and data across different platforms like IoT/edge/fog/cloud through a reliable network with a remaining executable and useable, which is essential for maintaining performance and flexibility in IIoT/edge/fog environments [160]. Industrial cloud and fog service users, in particular, require high portability to enable the migration of IIoT applications to alternative resource providers or computing devices in order to improve key Quality of Service (QoS) parameters such as cost efficiency, latency reduction, and operational flexibility [161]. In the real world, it is difficult to achieve effective collaboration, seamless service delivery, and scalable expansion across distributed federated infrastructures without addressing these interoperability and portability aspects [162]. Future research should focus on the development of open standards, middleware solutions, and dynamic orchestration frameworks to ensure robust interoperability in practical deployments.

6.7 Fault tolerance in edge-to-cloud offloading

Implementing fault tolerance, resiliency, and data replication at edge-to-cloud computing is crucial for preserving the system state during node failures or occasional network connectivity issues. In a clustered system with multiple nodes transporting data between edge and cloud, malfunctioning edge nodes can cause permanent data loss [163]. Fault tolerance and resiliency are also essential to enable continuous operations and reliable data collection in highly distributed IIoT environments. Fault tolerance refers to the system's ability to continue running applications correctly despite failures in certain of its components, ensuring operational continuity during certain issues on the nodes or edge devices [164]. This is especially important for industrial processes like bank transactions that are unable to tolerate unexpected downtimes in servers or data losses [165]. On the other hand, resiliencywhile closely related to fault tolerance, emphasises the system's ability to recover quickly from disturbances or failures. Distributed edge-to-cloud computing offers a promising strategy to enhance both fault tolerance and resiliency by spreading computation and storage across multiple hierarchical layers, from local edge devices to centralized cloud servers [166, 167]. Future federated edge-to-cloud architectures should incorporate adaptive failure detection, resiliency, and redundancy techniques to strengthen system robustness and reliability.

7 Conclusion

Computation offloading is a fundamental strategy to meet the resource limitations of IoT systems and enable low-latency, scalable services in distributed environments, particularly within federated edge-to-cloud computing infrastructures. In this paper, we have conducted a thorough of a wide range of existing offloading solutions and subsequently proposed a novel taxonomy consisting of a diverse set of characteristics, ranging from rule-based to optimisation and intelligent machine learning-driven approaches, deemed essential for distributed computational offloading within the federated edge-to-cloud computing continuum.

Based on the obtained results from this review, we identified seven key challenges where current solutions show significant gaps. These areas include mobility, intelligent resource allocation, energy prediction, security, lack of datasets, Interoperability/portability, and fault tolerance across heterogeneous environments. Informed by these identified challenges, we believe that the emerging integration of AI-driven decision-making, distributed resource allocation, and federated trust models will play a crucial role in advancing the next generation of efficient and resilient edge-to-cloud offloading systems.

Author Contributions Jaber took the lead in all aspects of the work, including the initial conceptualisation, the design of the conceptual framework, and the writing of the entire manuscript. Amjad contributed to the taxonomy design and provided input for the Introduction, Research Methodology, and Taxonomy sections. Ahmed supported the conceptual framework design and conducted a comprehensive review of the manuscript. Xiaodong assisted with the conceptual framework and reviewed the full manuscript for coherence and technical accuracy.

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Data Availability The dataset used and analysed during the current study is available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval Not applicable.

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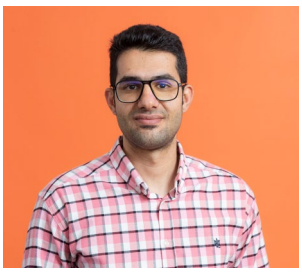
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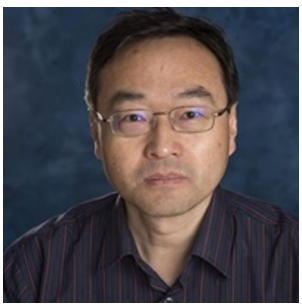
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